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Video diffusion generation: comprehensive review and open problems

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Abstract
Video generation has become an increasingly important component of AI-generated content (AIGC), owing to its rich semantic expressiveness and growing application potential. Among various generative paradigms, diffusion models have recently gained prominence due to their strong controllability, competitive visual quality, and compatibility with multimodal inputs. However, most existing surveys provide limited coverage of diffusion-based video generation, often lacking systematic analysis and comprehensive comparisons. To address this gap, this paper presents a thorough and structured review of diffusion models for video generation. We first outline the theoretical foundations and core architectures of diffusion models, and then the key design principles of representative methods for video generation were introduced. We propose a unified taxonomy that categorizes over two hundred methods, analyzing their key characteristics, strengths, and limitations. In addition, we compared the performance of classical methods and summarized commonly used datasets and evaluation metrics in this field for ease of model benchmarking and selection. Finally, we discuss open problems and future research directions, aiming to provide a valuable reference for both academic research and practical development.

Keywords Review · AI-generated content (AIGC) · Video diffusion models · Video generation

1 Introduction

At present, AI-generated content (AIGC) (Cao et al. 2023; Xu et al. 2024; Wu et al. 2023) has become one of the mainstream directions in artificial intelligence and computer vision (Voulodimos et al. 2018), achieving significant breakthroughs across various domains. AIGC enables intelligent functionalities such as speech synthesis (Zhang et al. 2023), image generation (Epstein et al. 2023; Chen et al. 2025), and video generation (Singer et al. 2022; Ho et al. 2022). Among these, video plays a particularly prominent role due to its rich information content and dynamic semantics (Han and Xi 2020) in today's digital era. Conse-

Extended author information available on the last page of the article quently, recent years have witnessed a surge in video generation algorithms driven by AIGC technologies. Notably, diffusion-based video generation models (Song et al. 2024; Croitoru et al. 2023; Yang et al. 2023) have emerged as a leading paradigm, gradually replacing traditional generative approaches (Chu et al. 2020; Pradhyumna et al. 2021; Yan et al. 2021). Originally designed for text-to-image synthesis (Saharia et al. 2022), diffusion models have been rapidly extended to the domain of video generation (Singer et al. 2022; Ho et al. 2022), supporting various tasks such as text-to-video (Wu et al. 2023), image-to-video (Ni et al. 2023), video-to-video translation (Yang et al. 2023), and multimodal generation (Ruan et al. 2023). Figure 1 illustrates the distribution of representative diffusion-based video generation methods across different task categories. Compared to traditional video generation algorithms (Chu et al. 2020; Pradhyumna et al. 2021; Yan et al. 2021), diffusion models offer superior output quality and generalization capabilities while significantly enhancing the flexibility and creativity of video generation.

Among various diffusion models for video generation, Denoising Diffusion Probabilistic Models (DDPMs) (Voleti et al. 2022) have demonstrated superior image quality and more stable training compared to traditional generative models such as GANs (Chu et al. 2020). Latent Diffusion Models (LDMs) (Khachatryan et al. 2023; Wang et al. 2023) further improve efficiency by mapping visual data from pixel space to a lower-dimensional latent space, significantly reducing the computational cost of video generation. Models like DALLE-2 and Stable Diffusion (Singer et al. 2022) incorporate multimodal architectures, such as Transformers (Yang et al. 2022), to translate natural language into the latent space of images. Recent advancements in video diffusion models (Ho et al. 2022; Voleti et al. 2022) enable multimodal video generation conditioned on text and images. Diffusion models effectively leverage their intrinsic advantages to generate high-quality videos with strong subject consistency and temporal coherence. However, challenges remain, particularly in generating long videos with high fidelity and ensuring consistency across multiple subjects. These issues present important directions for future research.

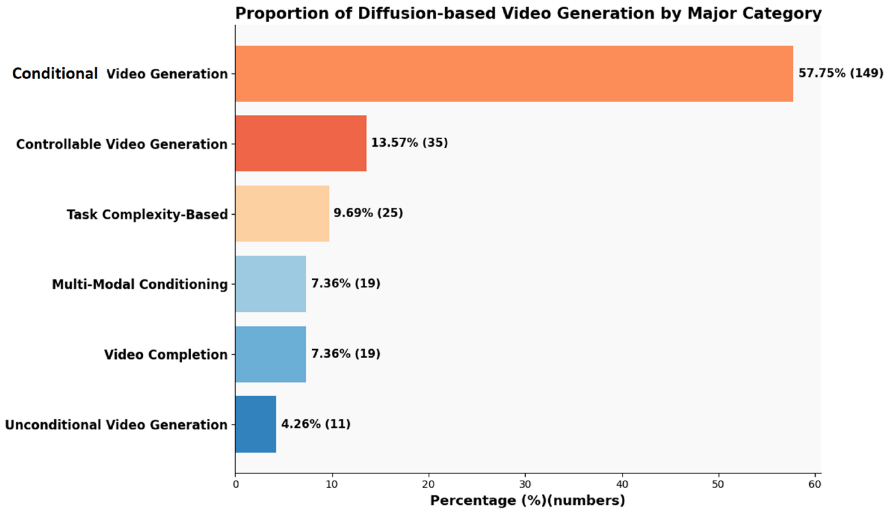

Fig. 1 Distribution of Research Methods in Different Categories of Diffusion-based Video Generation. The horizontal axis represents our six major classifications of models in this field, and the vertical axis represents the percentage of models under each classification

Recent studies have shown that numerous AIGC-related surveys (Xing et al. 2024; Li et al. 2024; Cho et al. 2024) have covered aspects of video generation. Specifically, Xing et al. (Xing et al. 2024) present a broad and systematic overview of video diffusion models, encompassing video generation, editing, and understanding. It provides a comprehensive summary of representative works, benchmarks, and settings in this domain, serving as an essential reference for researchers. Li et al. (Li et al. 2024) focuses on long video generation, detailing modal types, control signals, and key techniques such as Divide And Conquer and Temporal AutoRegressive paradigms. This survey highlights important challenges related to video quality and hardware considerations. Meanwhile, Cho et al. (Cho et al. 2024) emphasizes a broader range of video synthesis topics, including non-text-conditioned generation, text-video relationships beyond generation tasks, and analysis of survey articles and under-cited works.

Building upon these prior surveys, our work offers a more in-depth and structured analysis of diffusion-based video generation. We systematically review theoretical foundations, architectural mechanisms, and nuanced strengths and limitations across different methods. Furthermore, we highlight emerging directions such as graph-based diffusion models, which offer promising potential for enhancing scene coherence and complex event synthesis. By expanding upon the insights offered by these survey, this survey organizes the discussion around key theoretical foundations, representative networks, classification schemes, strengths and limitations, application scenarios, performance benchmarks, and future research challenges. Our main contributions are as follows:

(1) This paper provides a structured analysis of the theoretical foundations and core architectures of diffusion models. The Video Diffusion Model (VDM) (Ho et al. 2022) is selected as a representative method to exemplify the fundamental principles and design mechanisms in video diffusion generation. In addition, we briefly introduce the role of model fine-tuning and the integration of large-scale language models in video generation to help readers better grasp recent advancements in the field.
(2) This paper categorizes over two hundred diffusion video generation methods, analyzes their key characteristics, and discusses both their advantages and disadvantages.
(3) The paper also reviews commonly used datasets and evaluation metrics in video generation and presents comparative results of representative models to facilitate benchmarking and model selection.
(4) In light of current development trends, this paper systematically discusses future research directions and open problems of diffusion video generation while also outlining potential application scenarios to support further exploration and real-world deployment.

The rest of this paper is organized as follows. Section 2 introduces the development of diffusion models and their foundational network structures, providing readers with a clear understanding of the core principles and technical evolution. Section 3 systematically reviews representative and breakthrough video diffusion generation methods, categorizing them by generation type and discussing the strengths and limitations of each category. Section 4 summarizes commonly used datasets and evaluation metrics for video generation. Section 5 provides a comparative analysis and visual presentation of the performance of representative algorithms. Section 6 we discusses future research directions and open problems of video diffusion generation. Finally, Sect. 7 provides a summary of this paper.

2 Related work

In this section, we review the principles of diffusion models and their advanced variants, and summarize the conditional video generation algorithms built upon them. The overall structure of the diffusion model is shown in Fig. 2.

Diffusion models, introduced in recent years as a type of generative model, progressively evolve a simple data distribution into a complex one through reversible steps, enabling the creation of entirely new data samples based on training data. They are widely applied in image generation and video generation algorithms, and in the field of image synthesis, they have surpassed traditional Generative Adversarial Networks (GANs) with their powerful generative capabilities. Representative models, such as Denoising Diffusion Probabilistic Models (DDPM) (Ho et al. 2020), Score-Based Generative Models (SGM) (Song et al. 2020a), Denoising Diffusion Implicit Models (DDIM) (Song et al. 2020b), and Latent Diffusion Models (LDM) (Rombach et al. 2022), have made remarkable progress in the field of video generation in recent years. Let's introduce each of the diffusion models mentioned above one by one.

2.1 Denoising diffusion probabilistic models

By gradually adding noise to the data sample and generating new data through the reverse denoising process, DDPM progressively maps the data distribution to a simple Gaussian distribution (noise distribution) and then generates new data through reverse denoising. This model was first proposed by Ho et al. (2020) and has achieved excellent performance in image and video generation tasks. The generation process is divided into two stages: the forward process and the reverse process.

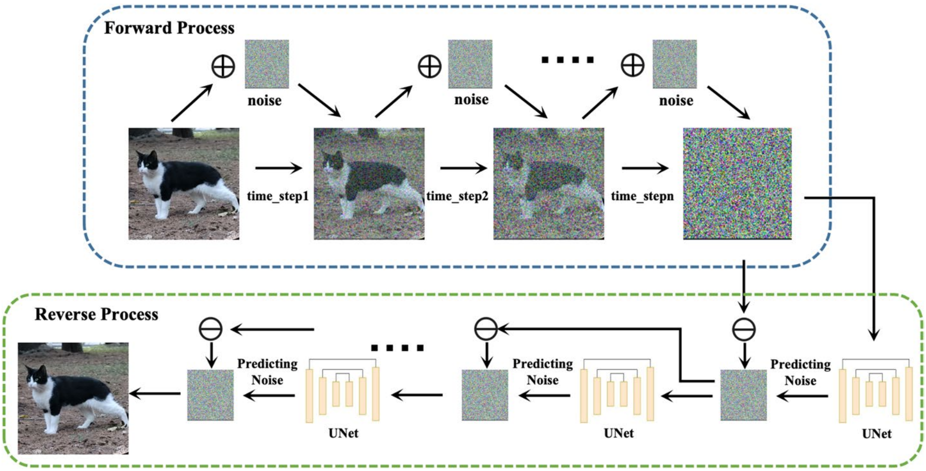

Fig. 2 The overall operational process of diffusion model

2.1.1 Forward process

The forward process, also known as the diffusion process, adds noise to the sample at each step until the data completely degenerates into Gaussian noise. Its essence is a Markov process of adding noise at a predefined time, which can be described using Eq. (1).

$$q\left(x_t \mid x_{t-1}\right)=\mathcal{N}\left(x_t ; \sqrt{1-\beta_t} x_{t-1}, \beta_t \mathrm{I}\right) \tag{1}$$

where $q(\cdot)$ is the conditional probability distribution of the forward process, β_t is the noise schedule corresponding to time step *t*, and $0<\beta_t<I$, $\mathcal{N}$ represents a normal distribution. At the same time, we can directly represent real data at any time t through a recursive Eq. (2):

$$\begin{cases} x_t=\sqrt{\bar{\alpha}_t} x_0+\sqrt{1-\bar{\alpha}_t} \epsilon, & \epsilon \sim \mathcal{N}(0, \mathrm{I}) \\ \text{s.t} \quad \bar{\alpha}_t=\prod_{s=1}^t\left(1-\beta_s\right) \end{cases} \tag{2}$$

where $\bar{\alpha}_t$ is the accumulated attenuation coefficient, ϵ is a standard Gaussian distribution.

2.1.2 Reverse process

The reverse process is to gradually remove noise and generate data from noise based on the reverse process of the forward process. The reverse process is also a Markov process, with the goal of estimating and removing noise at each step.

$$p_\theta(x_{t-1}|x_t)=\mathcal{N}(x_{t-1};\mu_\theta(x_t,t),\Sigma_\theta(x_t,t)) \tag{3}$$

where $p(\cdot)$ is the conditional probability distribution of the reverse process, $\mu_\theta(x_t,t)$ is the mean parameterized by neural networks (such as UNet) used to predict the denoised data at each step, and $\Sigma_\theta(x_t,t)$ is the covariance matrix, which can be fixed as a constant or learned using neural networks.

Overall, compared to GANs, DDPMs have better training stability and can generate high-quality images and videos. They are also more flexible and can control noise scheduling or incorporate external conditions (images or text) to guide the generation process (Dhariwal and Nichol 2021).

2.2 Score-based generative models

Based on the idea of score matching, SGM generate new data by learning the score function of the data distribution, which is the gradient of the logarithmic probability density of the data distribution. This method can be seen as a variant of the diffusion model, using a score function to guide the generation process. The forward process is similar to that of DDPMs.

$$x_t=x_0+\sigma_t\epsilon, \quad \epsilon\sim\mathcal{N}(0,\mathrm{I}) \tag{4}$$

where σ is the time-varying noise intensity, and ϵ is the standard Gaussian noise. Through Eq.(4) Add noise to gradually move the data away from the true distribution, ultimately becoming Gaussian noise. In the reverse process, SGM gradually removes noise by learning the fractional function of the data distribution, such as Eq.(5) As shown:

$$\nabla_{x_t} \log p(x_t) \tag{5}$$

Among them, t represents the time step, and the original data x_0 can be restored to the maximum extent by adjusting the current state x_t. The reverse process formula is shown in Eq.(6) as shown:

$$\frac{dx_t}{dt} = \frac{1}{2}\nabla_{x_t} \log p(x_t) \tag{6}$$

2.3 Denoising diffusion implicit models

As an improvement of DDPM, DDIM reduces the number of generation steps through a deterministic generation process while maintaining generation quality. Therefore, DDIM can generate high-quality data similar to DDPM while greatly reducing the number of generation steps.

It is still a Markov process, adding noise at each step t to gradually diffuse the real data into standard Gaussian noise. The key innovation of DDIM lies in the reverse process as shwon in Eq.(7).

$$\begin{aligned} x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \right) \\ + \sqrt{1-\bar{\alpha}_{t-1}} \cdot \epsilon_\theta(x_t, t) \end{aligned} \tag{7}$$

where $\epsilon_\theta(x_t, t)$ is the noise predicted by the model at time step t, $\bar{\alpha}_t$ is the cumulative noise scaling factor during the forward process. Unlike DDPM, the reverse process of DDIM is deterministic rather than based on random sampling. DDIM predicts the next state through a direct mapping, thereby reducing the number of generated steps.

2.4 Latent Diffusion Models

LDM is an efficient diffusion model, and the core idea of LDM is to compress image or video data into a low dimensional latent space, perform diffusion and denoising on the latent space, and then map the generated latent representation back to a high-dimensional pixel space through a decoder. This method combines the powerful generation capability of diffusion models with the efficient representation learning of autoencoders such as VAE.

2.4.1 Forward process

LDM diffuses in latent space, and the diffusion process is similar to traditional DDPM, as shown in Eq.(8), adding noise to the latent representation at each step until Gaussian noise is finally obtained.

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathrm{I}) \tag{8}$$

where z_0 is a latent representation obtained through an encoder, $\bar{\alpha}_t$ is the cumulative scaling factor during the diffusion process. ϵ is standard Gaussian noise.

2.4.2 Reverse process

In the reverse process in Eq.(9), LDM starts from Gaussian noise in the latent space, gradually removes the noise, and restores the original latent representation. Then, the latent representation is mapped back to the high-dimensional data space through a decoder to obtain the generated samples.

$$z_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(z_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(z_t, t) \right) + \sigma_t \epsilon \tag{9}$$

where α_t is the noise scaling factor for time step *t*, σ_t is a random term in the reverse denoising process, usually set according to noise scheduling. $\epsilon_\theta(z_t, t)$ is the noise predicted by the model at time step *t*.

2.5 Spatiotemporal extension for video generation

While this model provides an efficient framework for image generation in latent space, modeling the temporal dynamics in videos requires additional architectural considerations. Representative works such as Video Diffusion Models (VDM)(Ho et al. 2022) extend 2D UNet to 3D UNet structures and introduce spatiotemporal attention mechanisms to jointly capture spatial and temporal dependencies. As shown in Fig. 3, by decoupling spatial and temporal attention and leveraging relative position embeddings, these designs enhance temporal coherence across frames while maintaining computational efficiency, laying a solid foundation for later video diffusion models.

A key innovation in VDM is the use of reconstruction-guided sampling to improve conditional generation. This technique addresses the challenge of generating coherent, longer videos by adjusting the denoising model to include a gradient term based on the model's reconstruction of the conditioning data. The adjusted denoising model is defined as:

$$\tilde{x}_\theta(z_t) = \hat{x}_\theta(z_t) - w_r \frac{\alpha_t}{2} \nabla_{z_t} \left\| x^a - \hat{x}^a_\theta(z_t) \right\|_2^2 \tag{10}$$

where the $\tilde{x}_\theta(z_t)$ is the output of the standard denoising model, and x^a is the conditioning data (e.g., previous frames or low-resolution frames), the w_r is a weighting factor (typically $w_r > 1$ for improved sample quality). α_t is a coefficient related to the time step *t*.

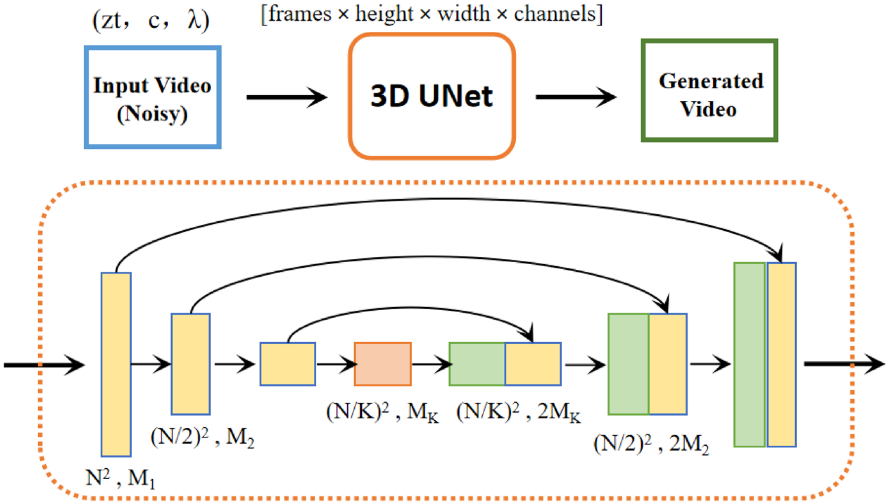

Fig. 3 The 3D UNet architecture and network of Video Diffusion Models (Ho et al. 2022). The model captures spatial-temporal features from high-resolution inputs (N^2, M_1) to compressed forms $(\frac{N}{2})^2, M_k$, with progressive upsampling ensuring both resolution and motion consistency

The architectural design of VDM enables high-resolution and temporally consistent video generation. It represents a significant advancement in extending diffusion models from static images to dynamic video sequences.

2.6 Model finetuning in video diffusion

Model finetuning has emerged as a crucial component in the diffusion-based video generation paradigm. Finetuning leverages pre-trained diffusion models by adapting them to specific tasks or domains. This process narrows the gap between general-purpose, large-scale models and high-performance, task-specific video generation systems. The key significance of model finetuning lies in its ability to substantially enhance video generation quality, including improved visual fidelity, temporal coherence, and alignment with conditioning inputs such as text, pose, or style, while maintaining high computational efficiency.

Finetuning methods, such as Parameter-Efficient Fine-Tuning (PEFT) (Lester et al. 2021), Low-Rank Adaptation (LoRA) (Hu et al. 2022), and prompt tuning (Lester et al. 2021), have been increasingly applied to enhance the adaptability and scalability of diffusion-based video generation systems. These techniques allow efficient updating of a subset of model parameters while leveraging the knowledge from large pre-trained models, thus reducing computational cost and data requirements.

Specifically, in the video diffusion field, models like Tune-A-Video (Wu et al. 2023), FreeInit (Wu et al. 2024), and FreeNoise (Qiu et al. 2023) have employed efficient finetuning strategies to adapt text-to-video models for high-quality video generation under limited computational resources. For instance, Tune-A-Video uses temporal linear adaptation modules to fine-tune large pre-trained text-to-image diffusion models for video synthesis, demonstrating the effectiveness of PEFT-like techniques in video generation tasks.

Other finetuning strategies include Adapter Layers (Houlsby et al. 2019), which insert trainable modules into pre-trained models; BitFit (Zaken et al. 2021), which only updates bias parameters; and meta-learning based finetuning such as Model-Agnostic Meta-Learning (MAML) (Finn et al. 2017), which enables models to quickly adapt to new video generation tasks. Incorporating these model finetuning strategies represents a promising direction for improving performance and flexibility across diverse video diffusion applications.

2.7 Vision-language foundation models in video generation

Recent advancements in vision-language foundation models (VLFMs), such as CLIP (Radford et al. 2021), BLIP (Li et al. 2022), BLIP-2 (Li et al. 2023), Flamingo (Alayrac et al. 2022), and PaLI (Chen et al. 2022), have greatly enhanced the integration of visual and textual modalities in generative tasks. These models, pre-trained on massive-scale image-text datasets, produce semantically rich, cross-modal representations that align visual content with natural language descriptions. These models capture both global and fine-grained semantics, enabling strong contextual understanding and robust cross-modal alignment.

In the context of diffusion-based video generation, VLFMs play an increasingly pivotal role. By incorporating pre-trained vision-language embeddings, models can effectively guide the generation process to produce videos that are not only visually realistic but also semantically aligned with input text. For example, cross-attention mechanisms in video diffusion models can seamlessly integrate BLIP-generated textual embeddings into the latent space, enabling precise control over content generation and enhancing temporal consistency.

In addition, Flamingo introduces a retrieval-augmented multimodal transformer architecture that extends vision-language modeling to handle long sequences and dynamic video data. PaLI further advances this by integrating multilingual text and diverse visual representations, supporting richer global context and accommodating complex user instructions. CLIP, while primarily designed for contrastive alignment, provides a robust foundation for conditioning diffusion models through its powerful image-text embedding capabilities, often used in zero-shot video understanding and retrieval.

Incorporating vision-language foundation models into diffusion-based video generation brings several key benefits. Firstly, it enhances the semantic fidelity of generated videos, ensuring that they closely align with complex and nuanced natural language prompts. Secondly, it improves the controllability of generation processes, enabling precise adjustments to aspects such as style, motion, and content. Lastly, it opens up possibilities for multi-modal and multi-lingual generation, which is crucial for addressing the challenges of global-scale video synthesis.

Despite these benefits, current diffusion-based video generation methods largely rely on simple text encoders or static embeddings, which limit the generative capacity in complex scenarios. The integration of advanced VLFMs provides a path towards more intelligent and human-aligned generation systems, enabling models to interpret nuanced textual instructions and generate temporally coherent, high-quality videos. This trend underscores the importance of incorporating VLFMs into future video diffusion frameworks, paving the way for next-generation text-to-video models capable of handling long-duration, multi-scene, and richly conditioned video synthesis tasks.

3 Classification of methods

In this section, we introduce a comprehensive classification of diffusion-based video generation methods, as well as the model characteristics, analysis of advantages and disadvantages, and future research directions of each classification, as shown in Table 1. This classification is organized according to multiple dimensions of current research. Specifically, we divide the methods into six primary categories: Task Complexity-Based, Multi-Modal Conditioning, Controllable Generation, Conditional Generation, Unconditional Generation, and Video Completion. Each category is further refined based on shared characteristics in training requirements, conditioning modalities, or generation targets. This structured taxonomy not only facilitates a clearer understanding of the research landscape but also highlights emerging trends and gaps for future exploration. Figure 4 visualizes the distribution of works under this classification. Below, we introduce the models corresponding to each category as organized in Table 1.

In addition, to better illustrate the evolution of diffusion-based video generation, we provide a timeline of milestone works covering the years 2020 to 2025 (see Fig. 5). This visual summary highlights key breakthroughs and influential models in the field, including foundational developments, architectural innovations, and representative works that have driven major progress in video quality, controllability, and realism. In particular, we emphasize the most influential and recommended works in each year using a yellow star ($\star$), guiding readers to the most impactful models for further study. These include early foundational diffusion models (DDPM, DDIM), the introduction of temporal and latent modeling in VDM and LVDM, the first action-aware architecture in AnimateDiff, and the recent breakthroughs such as Sora, which marks a leap toward large-scale end-to-end text-to-video generation.

3.1 Task complexity-based

In this part, we introduce a video generation classification method based on task complexity. This classification method mainly focuses on the duration and complexity of the generation task, dividing the model into training-based long video generation and training-free long video generation:

3.1.1 Long video generation (training-based)

The training-based long video generation methods requires large-scale pre-training or fine-tuning of specific tasks to generate high-quality long videos with temporal consistency. Usually, these methods have high computational costs, but they perform better in terms of video quality and detail control compared to other methods. The key characteristics of this category are summarized below.

- **Multi-stage generation mechanism:** Most methods adopt a stepwise, autoregressive strategy to generate long videos frame by frame, ensuring coherence by referencing previous frames. ViD-GPT (Gao et al. 2024) and StreamingT2V (Henschel et al. 2024) follow this approach, while FlexiFilm (Ouyang et al. 2024) introduces a temporal conditioner to enhance multimodal alignment.
- **Time consistency and long-term memory:** Ensuring temporal consistency is crucial.

Table 1 Overview of diffusion-based video generation

Task Complexity-Based	
Long Video (Training-based)	FreeNoise (Qiu et al. 2023), NUWA-XL (Yin et al. 2023), Gen-L-Video (Wang et al. 2023), VIDiff (Xing et al. 2023), StreamingT2V (Henschel et al. 2024), FlexiFilm (Ouyang et al. 2024), ViD-GPT (Gao et al. 2024), Multi-sentence (Feng et al. 2024), Cono (Wang et al. 2024), Video-Infinity (Tan et al. 2024), DiVE (Jiang et al. 2024), ARLON (Li et al. 2024), SCA-CRV (Yan et al. 2024), Owl-1 (Huang et al. 2024), Mind the Time (Wu et al. 2025), MaskFlow (Fuest et al. 2025), WorldDreamer (Wang et al. 2024), WM-ML (Liu et al. 2025)
Long Video (Training-free)	FIFO-Diffusion (Kim et al. 2024), FreeLong (Lu et al. 2024), DiTCtrl (Cai et al. 2024), ConFiner (Li et al. 2024), TVG (Zhang et al. 2024), TFL (Ma et al. 2025), Ouroboros-Diffusion (Chen et al. 2025)
Multi-Modal Conditioning	
Mutil-modal	VideoComposer (Wang et al. 2023), MovieFactory (Zhu et al. 2023), NExT-GPT (Wu et al. 2024), MM-Diffusion (Ruan et al. 2023), PEEKABOO (Jain et al. 2024), CMMD (Yang et al. 2024), Fine-grained (Huang et al. 2023), GPT4Video (Wang et al. 2024), Panacea (Wen et al. 2023), SparseCtrl (Guo et al. 2023), AnimateLCM (Wang et al. 2024), ActAnywhere (Pan et al. 2024), Custom-Video (Wang et al. 2024), MoonShot (Zhang et al. 2024), UniCtrl (Xia et al. 2024), Magic-Me (Ma et al. 2024), InteractiveVideo (Zhang et al. 2024), Direct-a-Video (Yang et al. 2024), Boximator (Wang et al. 2024)
Controllable Video Generation	
Multi-View and Camera Control-Based	InterDy (Akkerman et al. 2024), OmniDrag (Li et al. 2024), SynCamMaster (Bai et al. 2024), CamI2V (Zheng et al. 2024), Cavia (Zheng et al. 2024), Training-free Camera Control (Hou et al. 2024), ObjCtrl-2.5D (Wang et al. 2024), v3d (Chen et al. 2024), ReCamMaster (Bai et al. (2025))
Animation Generation-Based	AnimateAnyone (Hu et al. 2024), AnimateAnything (Dai et al. 2023), FlipSketch (Bandyopadhyay and Song 2024), X-Portrait (Xie et al. 2024), LivePortrait (Guo et al. 2024), ExAvatar (Moon et al. 2024)
Motion Trajectory Control-Based	DragNUWA (Chen et al. 2023), Generative Inbetweening (Fu et al. 2024a), 3DTrajMaster (Fu et al. 2024b), Motion Prompting (Geng et al. 2024), FreeTraj (Qiu et al. 2024), MotionClone (Ling et al. 2024), TrailBlazer (Ma et al. 2024), Motion-Zero (Chen et al. 2024)
Model Optimization-Based	EasyControl (Wang et al. 2024a), FRAMER (Wang et al. 2024b), Tora (Zhang et al. 2024), HumanVid (Wang et al. 2024), Cinemo (Ma et al. 2024), Still-Moving (Chefer et al. 2024), Image Conductor (Li et al. 2024), Identifying Conditional Image Leakage (Zhao et al. 2024), MimicMotion (Zhang et al. 2024), MOFA-Video (Niu et al. 2024), Champ (Zhu et al. 2024), ReCapture (Zhang et al. 2024), Motion-Ctrl (Wang et al. 2024)
Conditional Video Generation	

Table 1 (continued)

Task Complexity-Based	
Text-to-Video(Training-based)	VDM (Ho et al. 2022), Imagen Video (Ho et al. 2022), Make-A-Video (Singer et al. 2022), MagicVideo (Zhou et al. 2022), LVDM (He et al. 2022), Cogvideo (Hong et al. 2022), PYoCo (Ge et al. 2023), AnimateDiff (Guo et al. 2023), VidRD (Gu et al. 2023), Fusionframes (Arkhipkin et al. 2023), Latent-shift (An et al. 2023), Videodirectorgpt (Lin et al. 2023), Text2Performer (Jiang et al. 2023), NUWA-XL (Yin et al. 2023), DSDN (Liu et al. 2023), ModelScope (Wang et al. 2023), VideoGEN (Li et al. 2023), Video Adapter (Yang et al. 2023), SVD (Blattmann et al. 2023), VersVideo (Xiang et al. 2023), Control-A-Video (Chen et al. 2024), Dysen-VDM (Fei et al. 2024), Simda (Xing et al. 2024), LAVIE (Wang et al. 2024), W.A.L.T (Gupta et al. 2024), GridDiff (Lee et al. 2024), FlexiFilm (Ouyang et al. 2024), PixelDance (Zeng et al. 2024), Cogvideox (Yang et al. 2024), EMU-video (Girdhar et al. 2024), RF (Long et al. 2024), CMD (Yu et al. 2024), Snap Video (Menapace et al. 2024), GenTron (Chen et al. 2024), Lumiere (Bar-Tal et al. 2024), ART•V (Wang et al. 2024), Vlogger (Zhuang et al. 2024), Hierarchical Spatio-temporal Decoupling (Qing et al. 2024), DiffPerformer (Wang et al. 2024), Generative Rendering (Cai et al. 2024), Hierarchical Patch-wise (Skorokhodov et al. 2024), Grid Diffusion Models (Lee et al. 2024), MotionDirector (Zhao et al. 2024a), MagDiff (Zhao et al. 2024b), MoVideo (Liang et al. 2024), HARIVO (Kwon et al. 2024), EvalCrafter (Liu et al. 2024), Make Pixels Dance (Zeng et al. 2024), Mind the Time (Wu et al. 2025), Show-1 (Zhang et al. 2024), DrivingDiffusion (Li et al. 2024), E2HQV (Qu et al. 2024), WM-ML (Liu et al. 2025), WorldDreamer (Wang et al. 2024), MagicVideo-V2 (Wang et al. 2024), Mora (Yuan et al. 2024), FancyVideo (Feng et al. 2024), Factorized-Dreamer (Yang et al. 2024), DirectorLLM (Song et al. 2024), Sora (Cho et al. 2024), HunyuanVideo(Kong et al. 2024), BlobGEN-Vid (Feng et al. 2025), VideoFactory (Wang et al. 2025)
Text-to-Video(Training-free)	Text2video-Zero (Khachatryan et al. 2023), Tune-A-Video (Wu et al. 2023), FreeNoise (Qiu et al. 2023), FlowZero (Lu et al. 2023), AdaDiff (Zhang et al. 2023), DiffSynth (Duan et al. 2024), ConditionVideo (Peng et al. 2024), GPT4Motion (Lv et al. 2024), F^3 -Pruning (Su et al. 2024), FreeInit (Wu et al. 2024), TRAILBLAZER (Ma et al. 2024), VideoElevator (Zhang et al. 2024), FIFO-Diffusion (Kim et al. 2024), Mevg (Oh et al. 2024), Bivdiff (Shi et al. 2024), A Recipe for Scaling (Wang et al. 2024), Fine-gained Zero-shot (Chen et al. 2024), Broadway (Bu et al. 2024)
Pose-guided	DreaMoving (Feng et al. 2023), AnimateAnyone (Hu et al. 2024), MagicDance (Chang et al. 2023), Dancing Avatar (Qin et al. 2023), DreamPose (Karras et al. 2023), StableAnimator (Liu et al. 2024), MOFA-Video (Niu et al. 2024), MimicMotion (Zhang et al. 2024), Champ (Zhu et al. 2024), Action Reimagined (Wang et al. 2024), Do You Guys Want to Dance (Xu et al. 2024a), MagicAnimate (Xu et al. 2024b), DisCo (Wang et al. 2024), Follow Your Pose (Ma et al. 2024), SG-I2V (Namekata et al. 2025)
Motion-guided	DragNUWA (Yin et al. 2023), Customizing Motion (Materzynska et al. 2023), Lamd (Hu et al. 2023), AnimateAnything (Dai et al. 2023), MCDiff (Chen et al. 2024), MotionClone (Ling et al. 2024), MotionCtrl (Wang et al. 2024), Tora (Zhang et al. 2024), MOFA-Video (Niu et al. 2024), Champ (Zhu et al. 2024), Motion-I2V (Shi et al. 2024), Motion-Zero (Chen et al. 2024), Pix2Gif (Kandala et al. 2024)

Table 1 (continued)

Task Complexity-Based	
Image-guided	NUWA-Infinity (Wu et al. 2022), I2vgen-xl (Zhang et al. 2023), Make-It-4D (Shen et al. 2023), LaMD (Hu et al. 2023), Conditional I2V (Ni et al. 2023), SparseCtrl (Guo et al. 2023), Identity-Preserving (Yuan et al. 2024), PhysGen (Liu et al. 2024), Ti2v-Zero (Ni et al. 2024), TimeNoise (Zhao et al. 2024), Noise Rectification (Li et al. 2024), AtomoVideo (Gong et al. 2024), Animated Stickers (Yan et al. 2024), Consisti2v (Ren et al. 2024), I2v-adapter (Guo et al. 2024), LivePhoto (Chen et al. 2024), VideoBooth (Jiang et al. 2024), Decouple Content and Motion (Shen et al. 2024), Generative Image Dynamics (Li et al. 2024), Follow-Your-Click (Ma et al. 2024), ID-Animator (He et al. 2024), MegActor-Σ (Yang et al. 2024), LeviTor (Wang et al. 2024), ControlNeXt (Peng et al. 2025), Consistent Human Image and Video Generation (Cao et al. 2024), DreamVideo (Wang et al. 2025)
Audio/Sound-guided	Diverse-AVGen (Yariv et al. 2023), tpos (Jeong et al. 2023), AA-Diff (Lee et al. 2023), DiffAligner (Xing et al. 2024), TAVGBench (Mao et al. 2024), Draw-an-Audio (Yang et al. 2024), Tell-What-You-Hear (Liu et al. 2024), AV-Link (Haji-Ali et al. 2024), Hallo (Xu et al. 2024), Context-aware (Xuanyuan et al. 2024), EMO (Tian et al. 2024), AGAV-Rater (Cao et al. 2025), UniForm (Zhao et al. 2025), MusicInfuser (Hong et al. 2025)
Brain-guided	Cinematic Mindscapes (Chen et al. 2023), NeuroCine (Sun et al. 2024)
Depth-guided	Animate-A-Story (He et al. 2023), StableV2V (Liu et al. 2024), Make-Your-Video (Xing et al. 2024)
Unconditional Video Generation	
UNet-based	VIDM (He et al. 2022), VPDinPL (Yu et al. 2023), GD-VDM (Lapid et al. 2023), LEO (Liang et al. 2023), Hybrid VDM (Kim et al. 2024)
Transformer-based	TGV-TST (Ge et al. 2022), VDT (Lu et al. 2023), RTM-VQGAN-Trans (Liu et al. 2024), Latte (Ma et al. 2024), MegActor-Σ (Yang et al. 2024), WF-VAE (Li et al. 2024)
Video Completion	
Enhancement and Restoration	CaDM (Zhou et al. 2022), Look Ma, No Hands! (Chang et al. 2023), AVID (Zhang et al. 2023), Motion-Guided Latent Diffusion (Yang et al. 2023), LDMVFI (Danier et al. 2023), Upscale-A-Video (Zhou et al. 2023), Inflation with Diffusion (Yuan et al. 2024), Towards Language-Driven Video Inpainting (Wu et al. 2024), ReconX (Liu et al. 2024)
Video Prediction	MaskViT (Gupta et al. 2022), RaMViD (Höppe et al. 2022), McVd (Voleti et al. 2022), Diffusion Probabilistic Modeling for Video Generation (Yang et al. 2022), Flexible Diffusion Modeling of Long Videos (Harvey et al. 2022), LG-CVD (Yang et al. 2023), Seer (Gu et al. 2023), Control-A-Video (Chen et al. 2024), AID (Xing et al. 2024), STDiff (Ye and Bilodeau 2024)

StreamingT2V (Henschel et al. 2024) uses a Conditional Attention Module (CAM) to reference prior segments, while ViD-GPT (Gao et al. 2024) maintains coherence via unidirectional temporal attention based on preceding frames.

- **Detail enhancement and multimodal support:** To improve visual quality, methods like FlexiFilm (Ouyang et al. 2024) apply resampling to reduce detail loss, and Multi-Sentence (Feng et al. 2024) enhances diversity by combining video editing with text-driven clip generation.

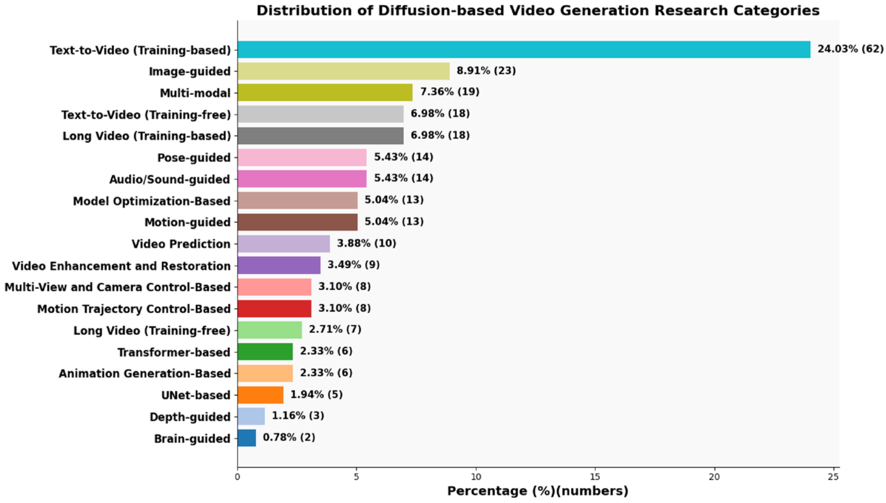

Fig. 4 Proportion of diffusion-based video generation papers in different categories

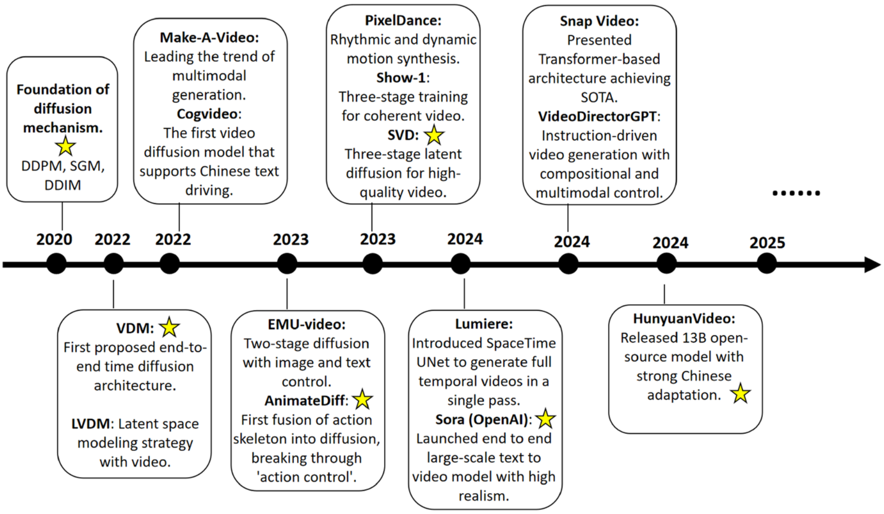

Fig. 5 The timeline showcases representative models and their contributions, year by year. Major milestones include the introduction of diffusion mechanisms (DDPM, DDIM), early video-specific adaptations (VDM, LVDM), multimodal fusion (Make-A-Video, AnimateDiff), and state-of-the-art end-to-end systems (Sora, HunyuanVideo). Yellow stars (⋆) highlight the most recommended and impactful works of each year

- **Optimization of computing resources and efficient inference:** Long, high-resolution videos demand substantial resources. ViD-GPT (Gao et al. 2024) reduces redundancy via KV caching to speed up inference, while Video Infinity (Tan et al. 2024) adopts distributed inference across GPUs to accelerate large-scale video generation.

3.1.2 Long video generation (training-free)

Training-free long video generation methods aim to synthesize extended video sequences without the need for task-specific pretraining or fine-tuning. These approaches typically leverage pre-trained image or short-video generation models and adapt them through prompt engineering, adapter modules, or lightweight optimization strategies. While they generally reduce computational overhead and enhance flexibility, challenges remain in achieving consistent temporal coherence and scene continuity over long durations. Below, we summarize the key characteristics of this method.

- **Efficient generation without training:** Training-free methods leverage pre-trained short-video models to extend to long video generation without additional training. FreeLong (Lu et al. 2024) reduces resource costs via frequency balancing, TVG (Zhang et al. 2024) applies Gaussian Process Regression for smooth transitions, and FIFO Diffusion (Kim et al. 2024) enables infinite generation with limited memory via a queue-based mechanism.
- **Frequency-domain temporal consistency:** These methods enhance consistency by optimizing in the frequency domain. FreeLong fuses global low-frequency and local high-frequency features to ensure visual balance without retraining.
- **Flexible inference process:** Training-free approaches allow flexible content control. FreeLong supports multi-segment prompts generation with seamless transitions, and TVG enhances transition naturalness through interpolation control and frequency fusion.

3.1.3 Analysis of the advantages and disadvantages

In this section, we present a structured analysis of the advantages and limitations of diffusion-based long video generation methods. Below, we explain the details of these advantages and disadvantages.

•**Advantages:**

Temporal Coherence: Both training-based and training-free methods strive to maintain consistent motion and visual flow over long durations. Approaches like StreamingT2V and FreeLong enhance frame continuity through temporal conditioning and frequency-aware design.

Flexible Architectures: Training-based models (e.g., FlexiFilm, Video-Infinity) adopt hierarchical or staged generation with time resampling and memory-aware modules to better handle long-range dependencies, while training-free models like FIFO-Diffusion extend short video capabilities without retraining.

Efficiency and Scalability: Training-free methods significantly reduce resource usage by reusing pre-trained models, enabling practical generation of long sequences at lower cost, with structures like causal queues and fast inference modules.

Multimodal and Fine-Grained Control: Training-based approaches offer stronger fidelity and fine control by integrating complex input conditions (text, image, keyframes), enabling accurate content generation over extended temporal spans.

•**Disadvantages:**

High Resource Requirements: Training-based long video generation consumes large GPU memory and time, with growing difficulty as video length increases. Distributed training is often necessary (e.g., Video-Infinity).

Suboptimal Quality in Training-Free Models: While efficient, training-free approaches usually lack fine-tuning mechanisms, leading to lower realism and weaker temporal alignment in complex scenes.

Limited Generalization: Some models are optimized for specific domains (e.g., animation or simulation videos) and may overfit or underperform when transferred to open-domain video generation.

Slow Optimization and Deployment: Training large models for long video generation is time-consuming and sensitive to hyperparameter tuning, which slows down development cycles and scalability across datasets.

3.2 Multi-modal conditioning

Multimodal video generation has rapidly advanced in recent years, enabling the synthesis of coherent and semantically rich videos by integrating conditions such as text, audio, images, and other modalities. Below we present the significant characteristics of multimodal video generation to reflect the technological evolution and diverse applications in this field.

- **Multimodal Conditioning:** Multimodal models integrate inputs like text, images, audio, or sketches to enable expressive and flexible video generation. Works such as Moonshot (Zhang et al. 2024) and InteractiveVideo (Zhang et al. 2024) showcase how diverse modalities and interactive prompts enhance customization.
- **Controllability and Customization:** Fine-grained control over motion, layout, and camera dynamics is key for personalized outputs. Direct-a-Video (Yang et al. 2024) and VideoComposer (Wang et al. 2023) offer spatial-temporal control mechanisms for precise content manipulation.
- **Temporal Consistency:** Maintaining frame-to-frame coherence is essential. Methods like UniCtrl (Xia et al. 2024), MM-Diffusion (Ruan et al. 2023), and VideoComposer introduce unified attention and motion-guided modules to enhance temporal stability.
- **Cross-Modal Alignment:** Ensuring semantic alignment across modalities is crucial. MM-Diffusion and Moonshot achieve this through attention-based mechanisms that align audio, text, and visual inputs, improving consistency and fidelity.
- **Unified Frameworks:** General-purpose models such as NExT-GPT (Wu et al. 2024) aim to unify diverse modalities within one architecture, enabling any-to-any generation via LLM-based adaptors and diffusion decoders.
- **User-Centric Interaction:** Modern systems like InteractiveVideo and MovieFactory focus on intuitive interfaces and language-driven pipelines, lowering the barrier for non-experts and enabling more creative, interactive video generation.

3.2.1 Analysis of advantages and disadvantages

In this part, we respectively introduce the advantages and disadvantages of mutilmodal video generation based on diffusion model. Below, we explain the details of these advantages and disadvantages.

•**Advantages:**

High-Quality Video Generation: Multimodal diffusion models can produce high-resolution, temporally consistent videos with rich details, meeting the increasing demand for realistic video synthesis. For instance, MovieFactory (Zhu et al. 2023) achieves cinematic-level quality with seamless transitions.

Flexible Input Modalities: These models support a wide range of input modalities, such as text, images, and audio, allowing users to generate videos based on diverse prompts. Examples include Direct-a-Video (Yang et al. 2024) and NExT-GPT (Wu et al. 2024), which enable customized video creation by separately controlling object motion and camera movement.

Enhanced Controllability: Fine-grained control mechanisms, such as those in MovieFactory and VideoComposer (Wang et al. 2023), empower users to adjust specific elements like object appearances, spatial layouts, and temporal dynamics, enhancing the precision of video synthesis.

Cross-Modal Alignment: By leveraging advanced alignment mechanisms (e.g., cross-modal attention), these models ensure synchronization and consistency across different modalities, such as video and audio, as demonstrated in MM-Diffusion (Ruan et al. 2023) and CMMD (Yang et al. 2024).

•**Disadvantages**:

Computational Complexity: The high computational demands of handling multiple modalities and maintaining video quality can result in significant training and inference time, especially for large-scale models like MovieFactory (Zhu et al. 2023).

Scalability Issues: Although modular approaches (e.g., NExT-GPT (Wu et al. 2024)) address scalability, performance may degrade when adding more modalities, primarily if significant differences exist between them.

Temporal Consistency Challenges: Despite efforts to enhance temporal consistency (e.g., MM-Diffusion (Ruan et al. 2023), VideoComposer (Wang et al. 2023)), complex dynamic scenes with rapid movements or multiple objects can still result in frame inconsistencies or distortions.

Dependency on Alignment Mechanisms: The quality of multimodal alignment is critical. Inadequate alignment can lead to inconsistencies between modalities, such as mismatches between audio and video or poor synchronization.

3.3 Controllable video generation

Controllable video generation enables fine-grained manipulation over video synthesis, allowing users to dictate motion, style, structure, and character identity through various control mechanisms. This field has rapidly evolved with advancements in generative models and diffusion-based architectures. Below, we analyze four major branches in controllable video generation, each addressing a unique aspect of control and customization.

3.3.1 Multi-view and camera control-based video generation

This class of methods enables the controllable generation of videos with flexible camera viewpoints and trajectories, supporting 2.5D/3D scene understanding and cinematic motion synthesis. The following are the key characteristics of this method.

- **Camera Motion Control:** Methods such as SynCamMaster (Bai et al. 2024) and CamI2V (Zheng et al. 2024) allow users to manipulate camera motion (e.g., pan, tilt, zoom) during generation, enabling realistic scene navigation and storytelling. ReCamMaster (Bai et al. 2025) extends this capability by introducing a camera-controlled generative video re-rendering framework that leverages pre-trained text-to-video models to reproduce dynamic scenes of an input video under novel camera trajectories, effectively bridging existing video and new camera path synthesis.
- **3D-Aware Scene Modeling:** Models like v3d (Chen et al. 2024) and ObjCtrl-2.5D (Wang et al. 2024) integrate 3D representations or geometry-aware controls to synthesize spatially consistent multi-view sequences.
- **Training-Free and Intuitive Interaction:** Approaches like Training-Free Camera Control (Hou et al. 2024) and OmniDrag (Li et al. 2024) support intuitive drag-based manipulation and viewpoint adjustment without retraining, improving user controllability and efficiency.
- **Object-Camera Dynamics:** InterDy (Akkerman et al. 2024) and Cavia (Zheng et al. 2024) jointly modal object and camera interactions to produce dynamic and coherent video outputs under complex scene transformations.

3.3.2 Animation generation-based video synthesis

These methods focus on animating static characters or objects with controllable motion, often used for portrait animation, sketch-to-video, or avatar-based generation.

- **Generic Object Animation:** AnimateAnything (Dai et al. 2023) and AnimateAnyone (Hu et al. 2024) support universal animation of arbitrary inputs, enabling broad applicability across human bodies, animals, and objects.
- **Portrait and Face Animation:** X-Portrait (Xie et al. 2024), LivePortrait (Guo et al. 2024), and ExAvatar (Moon et al. 2024) specializes in realistic facial animation, preserving identity and expression through audio, pose, or sketch conditioning.
- **Sketch-Based and Style-Driven Animation:** FlipSketch (Bandyopadhyay and Song 2024) enables frame-consistent animation from hand-drawn sketches, offering controllable style transfer and keyframe interpolation for creative animation workflows.

3.3.3 Motion trajectory control-based video generation

Motion trajectory control-based video generation focuses on synthesizing videos by explicitly guiding object or human movement along predefined trajectories, enabling precise control over spatial-temporal dynamics in both 2D and 3D space. The following outlines the key characteristics of this generation paradigm.

- **Custom Motion Paths:** Methods like Generative Inbetweening (Fu et al. 2024a) and 3DTrajMaster (Fu et al. 2024b) provide deep-learning-based interpolation between keyframes, ensuring natural motion transitions.
- **Motion-Conditioned Generation:** Approaches such as FreeTraj (Qiu et al. 2024) and Motion Prompting (Geng et al. 2024) incorporate motion priors or prompts to synthesize physically plausible trajectories with dynamic consistency.

- **Interactive Refinement:** Models like DragNUWA (Chen et al. 2023), TrailBlazer (Ma et al. 2024), and FRAMER (Wang et al. 2024b) support user interaction via text, key points, or bounding boxes, enabling fine-grained motion editing without additional retraining.

3.3.4 Model optimization-based video generation

Model optimization-based video generation emphasizes improving diffusion-based video generation's efficiency, scalability, and flexibility through lightweight architectures, training-free adaptation, and enhanced motion control. The following highlights the key characteristics of this class of approaches.

- **Efficient Fine-Tuning:** EasyControl (Wang et al. 2024a) enables fast adaptation with parameter-efficient tuning and lightweight design, reducing resource demands for real-time and scalable generation.
- **Style Adaptation and Personalization:** Tora (Zhang et al. 2024) and Cinemo (Ma et al. 2024) improve cross-domain generalization and temporal coherence by learning robust, style-aware embeddings.
- **Motion Consistency and Artifact Reduction:** Identifying Conditional Image Leakage (Zhao et al. 2024) and Image Conductor (Li et al. 2024) reduce motion artifacts and enhance realism by addressing over-reliance on conditional inputs and enabling better control over dynamic scenes.
- **Novel View Synthesis:** ReCapture (Zhang et al. 2024) and MOFA-Video (Niu et al. 2024) extend control to camera and scene-level trajectories, supporting cinematic motion planning and immersive video generation.

Each branch contributes to the broader field of controllable video generation, enhancing user interactivity, motion realism, and computational efficiency in modern video synthesis frameworks.

3.3.5 Analysis of the advantages and disadvantages

This part analyzes the key advantages and disadvantages of controllable video generation based on diffusion models. Below, we explain the details of these advantages and disadvantages.

•**Advantages:**

Fine-Grained Control: Controllable video generation enables precise manipulation of object motion, camera viewpoints, and animation styles. Models like ControlNeXt (Peng et al. 2025) and DirectorLLM (Song et al. 2024) allow users to guide video synthesis through textual, visual, and motion inputs, enhancing customization.

Diverse Control Mechanisms: Different approaches, such as condition-based (Control-A-Video (Chen et al. 2024), SparseCtrl (Guo et al. 2023)), multi-view camera control (OmniDrag (Li et al. 2024), CamI2V (Zheng et al. 2024)), and trajectory-guided motion (DragNUWA (Chen et al. 2023), TrailBlazer (Ma et al. 2024)), provide versatile methods to control video generation across various applications.

Enhanced Temporal Consistency: Advances in trajectory modeling (e.g., Motion-Zero (Chen et al. 2024), FreeTraj (Qiu et al. 2024)) and pose-guided animation (e.g., AnimateAnything (Dai et al. 2023), X-Portrait (Xie et al. 2024)) improve temporal coherence, reducing flickering and maintaining smooth transitions between frames.

Scalability Across Domains: Controllable video generation frameworks are adaptable to multiple domains, from animation (Animate Anyone (Hu et al. 2024), LivePortrait (Guo et al. 2024)) to cinematic video production (Cinemo (Ma et al. 2024), HumanVid (Wang et al. 2024)), expanding their applicability in content creation.

•**Disadvantages:**

High Computational Cost: Fine-grained control mechanisms require significant computational resources, particularly for models relying on diffusion processes (e.g., ControlNeXt (Peng et al. 2025), MotionCtrl (Wang et al. 2024)). Real-time inference remains challenging.

Limited Generalization to Open-Domain Videos: While trajectory-based and multi-view models perform well on structured datasets (e.g., Human3.6M), their ability to generalize to arbitrary open-domain videos remains limited (e.g., DragNUWA (Chen et al. 2023)).

Trade-off Between Control and Realism: Increased controllability may introduce artifacts or degrade natural motion, as observed in motion-conditioned models (e.g., Motion-Zero (Chen et al. 2024), MotionClone (Ling et al. 2024)). Overconstraining object trajectories can lead to unnatural deformations.

Complex User Input Requirements: Some methods (e.g., TrailBlazer (Ma et al. 2024), Control-A-Video (Chen et al. 2024)) require precise keyframe annotations or bounding box inputs, making them less accessible to users without prior expertise in motion design.

3.4 Conditional video generation

Conditional video generation enables explicit control over the video synthesis process using different conditioning inputs such as text, pose, motion trajectories, and external visual references. These methods allow for fine-grained manipulation of video content, ensuring alignment with user-specified conditions. Below, we analyze the key characteristics of various condition-based video generation approaches.

3.4.1 Text-to-video (training-based)

Text-to-video (training-based) generation methods rely on large-scale supervised training or fine-tuning to model the mapping from text prompts to coherent video sequences. These approaches often employ hierarchical latent diffusion frameworks, multi-stage generation pipelines, and sophisticated cross-modal alignment mechanisms to ensure high video fidelity and strong semantic consistency with the input text. The following summarizes the key characteristics of training-based text-to-video generation models.

- **High-Fidelity Video Synthesis:** High-resolution video generation methods such as Imagen Video (Ho et al. 2022), Make-A-Video (Singer et al. 2022), and LVDM (He et al. 2022) adopt cascaded or frame-insertion strategies to refine spatial quality and object details progressively. VideoGEN (Li et al. 2023) combines scripts and keyframes to enhance cross-shot consistency, while Show-1 (Zhang et al. 2024) integrates pixel-based and latent-based VDMs for better text alignment and high-resolution outputs.

- **Temporal Coherence and Motion Consistency:** Maintaining motion smoothness across frames is critical. Dysen-VDM (Fei et al. 2024) and LVDM (He et al. 2022) introduce scene-aware modeling and optical flow refinement. In contrast, VidRD (Gu et al. 2023) injects temporal layers into the decoder to strengthen long-range coherence without relying on separate prediction modules.
- **Modular and Efficient Generation:** ConFiner (Li et al. 2024) decouples video generation into structure control and spatiotemporal refinement, leveraging diffusion experts for each subtask to reduce computational cost while maintaining visual quality.
- **Structured Layout and Content Control:** VideoDirGPT (Lin et al. 2023) introduces GPT-based video planning, converting textual input into structured scene layouts and movement instructions. This enables controllable and coherent synthesis of complex, multi-entity scenes.
- **Compositional Modeling and Scene Adaptation:** VideoFactory (Wang et al. 2025), PYoCo (Ge et al. 2023), and W.A.L.T (Gupta et al. 2024) enhance spatial-temporal interactions through advanced attention mechanisms, video priors, and transformer-based encoders. CMD (Yu et al. 2024) disentangles content and motion latent spaces for faster and more controllable generation using pre-trained image diffusion backbones.
- **Scalability and Efficiency Improvements:** Scaling video generation models efficiently is a growing focus. Snap Video (Menapace et al. 2024) and GridDiff (Lee et al. 2024) reduce memory and training overhead via video-first and grid-based representations. GenTron (Chen et al. 2024) scales Transformer-based diffusion models to 3B parameters, achieving improved visual fidelity and motion-aware T2V synthesis.

3.4.2 Text-to-video (training-free)

Training-free methods have recently emerged as a lightweight and flexible alternative for video generation. These approaches typically adapt pre-trained text-to-image diffusion models to handle temporal dynamics, enabling zero-shot or few-shot video synthesis without expensive retraining. Next, we introduce the key characteristics of training-free text-to-video generation.

- **Fast Adaptation to New Prompts:** Text2video-Zero (Khachatryan et al. 2023), GPT-4Motion (Lv et al. 2024), and TrailBlazer (Ma et al. 2024) enable quick video generation from new prompts by leveraging pre-trained text-to-image models, eliminating the need for task-specific fine-tuning.
- **Temporal Consistency via Adaptive Inference:** FreeInit (Wu et al. 2024) improves motion smoothness by refining low-frequency components in initial latent noise. In contrast, FIFO-Diffusion (Kim et al. 2024) adopts a queue-based strategy to support infinite-length video generation with stable frame transitions.
- **Temporal Interpolation and Motion Synthesis:** FlowZero (Lu et al. 2023) uses LLM-generated dynamic scene syntax (DSS) to guide motion composition, avoiding direct optical flow. DiffSynth (Duan et al. 2024) reduces flickering by introducing latent deflickering and patch blending for smoother transitions.
- **Condition-Guided Temporal Control:** ConditionVideo (Peng et al. 2024) enhances motion accuracy and temporal coherence by combining a UNet branch with a control branch, using sparse bi-directional spatial-temporal attention and conditioning on exter-

nal inputs such as video and text.

- **Inference Acceleration and Resource Efficiency:** AdaDiff (Zhang et al. 2023) accelerates inference by adaptively adjusting denoising steps based on input complexity, while F^3-Pruning (Su et al. 2024) improves runtime by pruning redundant temporal attention modules without retraining.
- **Multi-Event Video Generation with Latent Optimization:** MEVG (Oh et al. 2024) enhances visual coherence across multiple events using a last-frame-aware latent refinement strategy, while a prompt generator improves semantic alignment through structured text conditioning.
- **Visual Quality and Temporal Diversity Enhancement:** VideoElevator (Zhang et al. 2024) boosts frame-level quality and structural consistency by decomposing generation into temporal refinement and spatial enhancement, leveraging T2I priors without extra training.

Training-free text-to-video generation offers a promising, fast, flexible, and resource-efficient video synthesis alternative. While current methods still face limitations in motion coherence and prompt adherence, ongoing research in adaptive frame interpolation, LLM-guided video synthesis, and lightweight diffusion modeling gradually bridges the gap with training-based approaches.

3.4.3 Pose-guided video generation

Pose-guided video generation leverages human pose sequences or key points as structural conditions to generate dynamic visual content. This approach enables control over human motion, body structure, and action dynamics, making it especially suitable for character animation, dance synthesis, and motion transfer applications. The following outlines the key characteristics of pose-guided video generation methods.

- **Pose-Conditioned Motion Transfer:** Methods like StableAnimator (Rings 2024), MagicAnimate (Xu et al. 2024b), AnimateAnyone (Hu et al. 2024), and Champ (Zhu et al. 2024) generate human videos guided by pose keypoints, enabling precise skeletal control through two-stage training and pose-aware encoders.
- **Style and Identity Preservation:** StableAnimator (Rings 2024), DreaMoving (Feng et al. 2023), and Champ (Zhu et al. 2024) preserve character appearance by combining identity-aware modules with pre-trained T2I backbones, maintaining visual consistency and text-editability across frames.
- **Expressive and Realistic Human Motion:** MOFA-Video (Niu et al. 2024), DisCo (Wang et al. 2024), and DreamPose (Karras et al. 2023) improve motion realism using pose-conditioned attention and dynamic embeddings, enabling smooth and expressive body articulation from sparse control signals.

3.4.4 Motion-guided video generation

Motion-guided video generation utilizes explicit motion trajectories or motion representations (e.g., optical flow, velocity maps) as conditions to control dynamic content in the generated video. This paradigm enables fine-grained manipulation of object motion and scene dynamics, providing greater realism and controllability in motion portrayal. Below, we summarize the key characteristics of motion-guided video generation methods.

- **Motion Trajectory Control:** Models like Tora (Zhang et al. 2024), Motion-I2V (Shi et al. 2024), and MOFA-Video (Niu et al. 2024) offer fine-grained trajectory-based motion control. Tora encodes motion paths via a 3D compression network to generate videos that precisely follow user-defined trajectories.
- **Dynamic Scene Adaptation:** Tora (Zhang et al. 2024) and FRAMER (Wang et al. 2024b) adapt motion synthesis to varying durations, resolutions, and layouts. These models maintain motion fidelity across diverse video settings through modules like Motion-Guidance Fuser.
- **Cross-Modal Motion Transfer:** MotionClone (Ling et al. 2024) and Motion-Zero (Chen et al. 2024) support motion transfer from reference videos or text prompts. MotionClone leverages sparse attention for lightweight transfer, while Motion-Zero introduces LLM-guided motion decomposition for region-specific control.

3.4.5 Image-guided video generation

Image-guided video generation focuses on synthesizing videos conditioned on single or multiple reference images, enabling the transfer of spatial content such as appearance, structure, or identity into temporally coherent sequences. These methods are particularly effective for tasks like image-to-video translation, video reenactment, and content preservation. Below, we outline the key characteristics of image-guided video generation approaches.

- **Identity and Appearance Preservation:** Maintaining a consistent identity is key in an image-driven generation. ConsisID (Yuan et al. 2024) preserves facial structure via frequency-aware control, while VideoBooth (Jiang et al. 2024) enhances subject coherence through coarse-to-fine prompt injection and attention guidance.
- **Motion Generation from Static Images:** To infer dynamics from a single image, PhysGen (Liu et al. 2024) introduces physics-based attributes, TI2V-Zero (Ni et al. 2024) combines text-image prompts with inversion, and LFDM (Li et al. 2024) generates flow-conditioned motion for spatial and temporal realism.
- **Controllable and Structured Generation:** AtomoVideo (Gong et al. 2024), ConsistI2V (Ren et al. 2024), and LaMD (Hu et al. 2023) provide structural and latent-level controllability using adapters, spatiotemporal attention, and prompt-guided latent motion control, enabling flexible and consistent generation.
- **Temporal Consistency for Long Videos:** NUWA-Infinity (Wu et al. 2022), DreamVideo (Wang et al. 2025), and Make-It-4D (Shen et al. 2023) improve long-term coherence through memory caching, subject-motion decoupling, and 4D scene modeling, achieving stable and realistic video synthesis over extended durations.
- **Robustness and Identity Leakage Mitigation:** Identifying Conditional Image Leakage (Zhao et al. 2024) and Noise Rectification (Li et al. 2024) reduce over-conditioning and enhance robustness via noise-aware scheduling and tuning-free correction strategies, improving generalization and detail preservation.

3.4.6 Audio/Sound-guided video generation

Audio/Sound-guided video generation leverages audio signals—such as speech, music, or environmental sounds, as conditioning inputs to control or synchronize the visual content in generated videos. This category emphasizes audio-visual alignment, semantic coherence, and expressive motion generation. Below, we highlight the key characteristics of audio/sound-guided video generation methods.

- **Audio-Driven Motion and Expression:** Models like Hallo (Xu et al. 2024), EMO (Tian et al. 2024), and TPoS (Jeong et al. 2023) generate expressive facial and gesture motion from speech using phoneme-to-motion mapping and emotion-aware conditioning.
- **Context-Aware Sound Synthesis:** Context-Aware Talking Face (Xuanyuan et al. 2024) aligns visual motion with beats, rhythm, and ambient sounds, enabling scene-aware and music-driven animations.
- **Audio-Visual Synchronization:** AADiff (Lee et al. 2023) and Hallo (Xu et al. 2024) use cross-modal attention and latent fusion to achieve precise lip-sync and temporal coherence between sound and frames.
- **Cross-Modal Alignment:** AV-Link (Haji-Ali et al. 2024) and Diverse-AVGen (Yariv et al. 2023) enhance semantic and temporal consistency between audio and video through aligned attention and AV-specific metrics.
- **Latent Mapping and Adaptation:** Uniform (Zhao et al. 2025) and AV-Link (Haji-Ali et al. 2024) build shared latent spaces or use adaptors for audio-to-visual conditioning without full model retraining.
- **Benchmarks and Evaluation:** TAVGBench (Mao et al. 2024) and AGAVQA (Cao et al. 2025) propose datasets and metrics (e.g., AVHScore, AGAV-Rater) for evaluating alignment and perceptual quality in audio-driven generation.

3.4.7 Brain-guided video generation

Brain-guided video generation utilizes brain activity signals—typically captured through modalities such as fMRI or EEG—to guide video synthesis. These methods aim to reconstruct or generate visual content aligned with cognitive or perceptual states, exploring the intersection of neuroscience and generative modeling. Below, we analyze the key characteristics of brain-guided video generation techniques.

- **Spatiotemporal Brain Decoding:** Models like Mind-Video (Chen et al. 2023) and NeuroCine (Sun et al. 2024) decode dynamic visual content from fMRI signals by leveraging masked brain modeling, temporal interpolation, and contrastive learning, enabling biologically aligned video reconstruction.
- **Diffusion-Based Generative Frameworks with Interpretability:** These methods incorporate temporally-aware diffusion models and attention-based decoding, achieving high-fidelity synthesis and neural interpretability through alignment with known brain structures.

3.4.8 Depth-guided video generation

Depth-guided video generation leverages depth information as a conditional input to guide the synthesis of temporally coherent and geometrically accurate videos. These methods can enhance spatial consistency, preserve structural details, and improve motion realism in generated sequences by incorporating depth cues. Below, we analyze the key characteristics of depth-guided video generation approaches.

- **Depth-guided Structural Control:** Models like StableV2V (Liu et al. 2024) and Animate-A-Story (He et al. 2023) use depth maps to guide motion and ensure shape-consistent generation across frames. This improves visual coherence and controllability by aligning motion with scene geometry.
- **Controllable and Personalized Generation:** Animate-A-Story (He et al. 2023) retrieves motion structures from reference clips and combines them with text prompts, These methods highlight the effectiveness of depth cues in enhancing spatiotemporal consistency and enabling scene-aware control in video generation.

3.4.9 Analysis of the advantages and disadvantages

This section summarizes the overall advantages and disadvantages of diffusion-based conditioned video generation methods. Below, we explain the details of these advantages and disadvantages.

•**Advantages:**

Cross-Modal and Fine-Grained Control: Conditioned generation enables precise control over spatial layout, temporal dynamics, and appearance by incorporating diverse inputs (e.g., text, pose sequences, depth maps, and audio signals). Representative models like VideoDirectorGPT (Lin et al. 2023) offer structured scene planning for multi-entity coordination, while MagicAnimate (Xu et al. 2024b) ensures identity preservation and fluid skeletal motion in human animations.

Temporal Coherence and Realism: Advanced models maintain smooth transitions and long-range temporal consistency through autoregressive or refinement strategies. MOFA-Video (Niu et al. 2024) and DreamPose (Karras et al. 2023) improve fluidity with dense motion modeling, while FreeInit (Wu et al. 2024) and NUWA-Infinity (Wu et al. 2022) leverage initialization or memory to extend generation length with reduced flicker and artifacts.

Scalability and Training Efficiency: Training-free paradigms like Text2video-Zero (Khachatryan et al. 2023), F^3-Pruning (Su et al. 2024), and AdaDiff (Zhang et al. 2023) enable fast deployment and reduce computation by leveraging pre-trained backbones and pruning techniques. Meanwhile, hybrid pipelines such as Show-1 (Zhang et al. 2024) and Grid-Diff (Lee et al. 2024) combine coarse-to-fine generation with modular structures, balancing visual quality and efficiency.

Domain Versatility and Personalization: Conditioned generation is widely applicable across diverse domains—from audio-driven talking faces (Xu et al. 2024; Tian et al. 2024) to brain-guided scene reconstruction (Chen et al. 2023; Sun et al. 2024). Models like Video-Booth (Jiang et al. 2024) and PhysGen (Liu et al. 2024) further support personalized outputs with controllable identity, emotion, or spatial layout, demonstrating flexibility in content customization.

•**Disadvantages:**

High Resource and Data Requirements: Many high-quality, training-based approaches (e.g., Imagen Video (Ho et al. 2022), Make-A-Video (Singer et al. 2022) demands large-scale training on multi-modal datasets, significant GPU resources, and fine-grained annotations (e.g., audio-visual or pose-aligned data), limiting their accessibility and scalability.

Generalization and Motion Limitations: Moreover, image/video-guided methods such as DreamVideo (Wang et al. 2025) may inherit static priors from their reference inputs, which hinders motion generalization and causes repetitive or unnatural dynamics.

Semantic Ambiguity and Alignment Issues: For audio-driven generation (Lee et al. 2023; Jeong et al. 2023), aligning acoustic features with visual semantics is inherently tricky, especially under noisy conditions. Brain-guided (Chen et al. 2023; Sun et al. 2024) and depth-guided (He et al. 2023; Xing et al. 2024) models further suffer from limited training data or poor input reliability, degrading generation quality.

Trade-off Between Control and Realism: Highly constrained synthesis may hinder realism. Over-conditioning—e.g., via overly rigid trajectories, bounding boxes, or depth constraints—can result in unnatural motion artifacts or visual inconsistencies, as observed in motion- and pose-driven settings (Chen et al. 2024; Zhu et al. 2024).

3.5 Unconditional video generation

Unconditional video generation synthesizes videos without explicit conditioning (e.g., text or motion cues) by autonomously learning temporal dynamics and visual patterns. Driven by diffusion models, recent approaches follow two main architectures: UNet-based and Transformer-based designs. In the following sections, we analyze these approaches' characteristics, pros and cons, and future research directions.

3.5.1 UNet-based video generation

In this part, we analyzed the characteristics of UNet-based video generation, and below are its details.

•**Spatial-Temporal Representation Learning:** UNet-based models capture multi-scale spatial-temporal features via encoder-decoder hierarchies. Hybrid Video Diffusion (Kim et al. 2024), GD-VDM (Lapid et al. 2023) integrate 2D/3D representations or depth priors to enhance realism and coherence.

•**Latent Diffusion for Efficient Generation:** Models like VPDinPL (Yu et al. 2023); VIDM (He et al. 2022); LEO (Liang et al. 2023) operate in latent space to reduce computational cost and accelerate generation. VPDinPL uses factorized latent encoding for long videos, while LEO disentangles motion and appearance to improve motion fidelity.

3.5.2 Transformer-based video generation

- **Temporal Dependency Modeling:** Transformer-based models capture long-range motion via self-attention. VDT (Lu et al. 2023) decouples spatial-temporal attention with unified masking, while Redefining Temporal Modeling (Liu et al. 2024) introduces vectorized timestep scheduling to improve flexibility and temporal stability.

- **Tokenized and Discrete Representation Learning:** VQ-based methods like VQGAN-Transformer (Ge et al. 2022) and Latte (Ma et al. 2024) tokenize video into discrete units for autoregressive prediction. Latte enhances efficiency with spatial-temporal decomposition, enabling scalable and controllable generation.

3.5.3 Analysis of the advantages and disadvantages

This section summarizes the advantages and disadvantages of diffusion-based **unconditional video generation** methods. Below, we explain the details of these advantages and disadvantages.

•**Advantages:**

Effective Spatialâ€"Temporal Representation Learning: UNet-based models such as VIDM (He et al. 2022) and LEO (Liang et al. 2023) leverage hierarchical convolution and latent-space modeling to generate high-resolution, temporally consistent videos while maintaining efficiency. Flow-based motion modules further improve motion realism and scene coherence.

Long-Range Temporal Dependency Modeling: Transformer-based methods like VDT (Lu et al. 2023); RTM-VQGAN-Trans (Liu et al. 2024); and Latte (Ma et al. 2024) utilize self-attention to capture long-range dependencies, supporting better motion continuity over extended frames and enhancing complex temporal structure modeling.

Flexible Conditioning and Token-Level Control: Transformers allow fine-grained control over sequence elements and enable seamless integration of text, structure, or modality information (Ge et al. 2022), which is difficult to achieve in CNN-based models.

High-Fidelity and Scalable Synthesis: By leveraging discrete latent representations (e.g., VQ tokens), transformer-based models improve semantic alignment and visual quality while offering different video lengths and resolutions scalability.

•**Disadvantages:**

Limited Long-Term Temporal Modeling in CNNs: UNet architectures often lack global receptive fields, making it difficult to capture long-term dependencies compared to self-attention-based models.

High Computational Overhead in Transformers: Transformer-based video generation introduces heavy memory and time consumption, especially when handling high-resolution or long-duration clips (Lu et al. 2023; Ma et al. 2024).

Slow Inference and Optimization Instability: Sequential attention and autoregressive mechanisms can significantly slow down inference compared to convolutional pipelines, and long sequence training may suffer from instability (Liu et al. 2024; Ma et al. 2024).

Architecture Rigidity in UNet Models: UNet-based pipelines offer less architectural flexibility for cross-modal tasks, and their reliance on latent compression may lead to loss of fine spatial details in long-term or complex scenarios.

3.6 Video completion

Video completion refers to generating or restoring missing, corrupted, or future frames in a video sequence. It includes video inpainting, super-resolution, denoising, frame interpolation, and predictive synthesis. These techniques are crucial in low-level restoration and high-level understanding of temporal dynamics, contributing to more robust and controllable video generation pipelines.

3.6.1 Enhancement and restoration

This task aims to improve the perceptual quality of degraded videos through diffusion-based techniques, focusing on super-resolution, denoising, inpainting, and temporal consistency.

- **Multimodal-Guided Restoration:** Methods like Language-Driven Video Inpainting (Wu et al. 2024) and AVID (Zhang et al. 2023) incorporate language or action cues to guide semantically-aware restoration, extending beyond visual-only enhancement.
- **Diffusion for Super-Resolution and Interpolation:** Inflation with Diffusion (Yuan et al. 2024) and LDMVFI (Danier et al. 2023) apply latent diffusion for high-quality upscaling and frame interpolation, improving perceptual smoothness and detail recovery.
- **Lightweight and High-Fidelity Upscaling:** Upscale-A-Video (Zhou et al. 2023) combines flow-guided propagation and text prompts for controllable, high-resolution video enhancement with temporal consistency.
- **User-Controllable Restoration:** CaDM (Zhou et al. 2022) and Look Ma, No Hands! (Chang et al. 2023) enable codec-aware and egocentric video restoration, introducing user or environment-driven control in restoration pipelines.

3.6.2 Video prediction

Video prediction focuses on forecasting future frames from a historical context, with applications in action anticipation, video completion, and autonomous driving. Diffusion-based methods enhance temporal coherence and prediction quality by leveraging structured conditioning, latent priors, and autoregressive modeling.

- **Future Frame Generation with Long-Term Consistency:** Models like STDiff (Ye and Bilodeau 2024) and MCVD (Voleti et al. 2022) employ stochastic motion modeling and masked training to improve continuity. AVID (Zhang et al. 2023) uses mid-frame attention to enable consistent text-guided inpainting across variable-length sequences.
- **Latent Diffusion for Predictive Motion Modeling:** LG-CVD (Yang et al. 2023) and RaMViD (Höppe et al. 2022) apply latent-space diffusion with context-aware modules for efficient, scalable prediction without relying on heavy 3D convolution.
- **Mask-Based and Flexible Conditioning:** MaskViT (Gupta et al. 2022) introduces masked modeling with temporal refinement for high-resolution prediction. Seer (Gu et al. 2023) adapts T2I models using spatiotemporal attention and decomposed text guidance.
- **Multimodal or Instructional Control:** Control-A-Video (Chen et al. 2024) combines text prompts and structural maps (e.g., edge, depth) with reward optimization to enhance alignment and controllability in future frame synthesis.

3.6.3 Analysis of the advantages and disadvantages

This section summarizes the advantages and disadvantages of diffusion-based **video completion** methods. Below, we explain the details of these advantages and disadvantages.

•Advantages:

Semantically-Aware Restoration and Enhancement: Methods such as Language-Driven Inpainting (Wu et al. 2024) and AVID (Zhang et al. 2023) utilize text or action cues for guided restoration, improving contextual accuracy and user alignment.

High-Quality Temporal Recovery: Diffusion-based techniques like LDMVFI (Danier et al. 2023) and Inflation with Diffusion (Yuan et al. 2024) produce superior spatial and temporal fidelity over traditional interpolation or super-resolution methods.

Lightweight and Scalable Design: Approaches such as Upscale-A-Video (Zhou et al. 2023) and CaDM (Zhou et al. 2022) adopt modular architectures suitable for deployment in practical scenarios with controllable resolution and robustness.

Long-Term Prediction with Flexible Control: Models like STDiff (Ye and Bilodeau 2024), MCVD (Voleti et al. 2022), and MaskViT (Gupta et al. 2022) support temporally stable forecasting with masked inputs. In contrast, Control-A-Video (Chen et al. 2024) leverages multimodal prompts (e.g., depth, edge, or text) to condition future predictions toward downstream task requirements.

•Disadvantages:

Temporal Instability in Dynamic Scenes: Despite progress, handling occlusion, fast motion, and long-term temporal consistency remain a challenge, often leading to drift or degradation (Voleti et al. 2022).

High Inference Cost: Most models require iterative denoising steps, introducing latency that hinders real-time applications (Ye and Bilodeau 2024).

Domain and Input Sensitivity: Performance may degrade across unseen conditions or misaligned prompts, especially in multimodal or controllable pipelines ((Chen et al. 2024), (Chang et al. 2023)).

Training Complexity and Generalization: Instruction-guided or hybrid models (e.g., Seer (Gu et al. 2023), Look Ma (Chang et al. 2023)) may suffer from instability and limited transferability without extensive domain adaptation.

4 Datasets and metrics

This section provides an overview of the commonly used datasets and evaluation metrics for video generation tasks, specifically focusing on video generation based on diffusion models.

4.1 Common datasets for video generation

The development of video generation models is closely tied to the availability of high-quality datasets. These datasets can be categorized into two main types: **caption-level** datasets, which provide video-text pairs to support tasks like text-to-video generation, and **category-level** datasets, which group videos by predefined action or semantic categories and are commonly used for unconditional or class-conditional generation. Below, we describe representative datasets from both categories.

4.1.1 Caption-level datasets

Caption-level datasets consist of video clips paired with descriptive textual captions, making them essential for text-to-video generation, captioning, and multimodal retrieval tasks.

We summarize major caption-level datasets in Table 2, covering early benchmarks such as MSR-VTT (Xu et al. 2016), large-scale resources like HowTo100M (Miech et al. 2019) and WebVid-10 M (Bain et al. 2021), and recent instruction-based datasets such as Señorita-2 M (Zi et al. 2025) and VideoUFO (Wang and Yang 2025). These datasets vary in

Table 2 Comparison of caption-level datasets for diffusion-based video generation

Dataset	Year	Clips	Resolution	Domain	Source
MSR-VTT (Xu et al. 2016)	2016	10K	240P	General	Manual
HowTo100M (Miech et al. 2019)	2019	136 M	240P–720P	Instructional	ASR (Miech et al. 2019)
WebVid-10 M (Bain et al. 2021)	2021	10.7M	360P	Open-domain	Alt-text
HD-VILA (Zellers 2022)	2022	103 M	720P	Open-domain	ASR
HD-VG-130 M (Wang et al. 2025)	2023	130 M	720P	Open-domain	Generated
InternVid (Wang et al. 2024)	2023	234 M	720P	Open-domain	Generated
CelebV-Text (Yu et al. 2023)	2023	70K	480P	Face-centric	Generated
Youku-mPLUG (Xu et al. 2023)	2023	10 M	720P	Chinese video	Generated
Panda-70 M (Chen et al. 2024)	2024	70.8M	720P	Open-domain	Generated
ChronoMagic-Pro (Yuan et al. 2024)	2024	460K	720P	Time-lapse	Manual
OpenVid-1 M (Nan et al. 2025)	2024	1 M	720P	General	Generated
Koala-36 M (Wang et al. 2024)	2024	36 M	512P	Open-domain	Refined + Gen
LVD-2 M (Xiong et al. 2024)	2024	2 M	512P	Long video	Manual
MovieBench (Wu et al. 2025)	2024	1K movies	1080P	Movie-level	Manual + Struct
VIVID-10 M (Hu et al. 2024)	2024	10 M	512P+	Local editing	Instruction-based
VideoBooth (Jiang et al. 2023)	2024	48.7K (train) / 650 (test)	–	Image+Text-conditioned	Filtered from WebVid-10 M
CustomVideo (Wang et al. 2024)	2024	68 pairs / 63 subjects	–	Multi-subject identity	Manual + Segmented
OpenHumanVid (Li et al. 2025)	2024	5 M	720P	Human-centric	Manual + Refined
Señorita-2 M (Zi et al. 2025)	2025	2 M	720P	Video editing	Expert-annotated
VideoUFO (Wang and Yang 2025)	2025	1 M	720P	User-focused	Instruction-based

supervision quality, domain coverage, and data source, supporting diverse generation tasks from open-domain synthesis to controllable editing.

4.1.2 Category-level datasets

As summarized in Table 3, category-level datasets span various domains, including human actions (e.g., UCF-101 (Soomro et al. 2012); Kinetics-400 (Kay et al. 2017); robotic tasks (e.g., BAIR (Ebert and et al. 2017); BridgeData (Walke et al. 2024), and autonomous driving (e.g., RDS (Liang and et al. 2022); Cityscapes (Cordts and et al. 2016). These datasets support the training and evaluation of video generation models across prediction, segmentation, and motion synthesis scenarios.

4.2 Metrics and benchmark for diffusion-based video generation

As summarized in Table 4, evaluation metrics for diffusion-based video generation can be broadly categorized into **image-level**, **video-level**, and increasingly, **specialized multimodal** metrics. Image-level metrics such as FID (Heusel et al. 2017), PSNR (Hore and Ziou 2010), SSIM (Wang et al. 2004), and CLIPSim (Radford et al. 2021) assess the visual quality and text alignment of individual frames.

Video-level metrics such as FVD (Unterthiner et al. 2019), KVD (Ryan et al. 2023), and Video IS (Wu et al. 2016) extend this evaluation to temporal aspects. Additional indicators like Frame Consistency (Wu et al. 2022), Motion Consistency (Liu et al. 2023; Tian et al. 2021), and Diversity Score (Saito et al. 2017) further evaluate frame smoothness, motion realism, and output diversity.

Recently, multimodal and perception-driven metrics have gained attention, including VideoCLIPSim (Fang et al. 2023) for video-text alignment, AV-Align (Haji-Ali et al. 2024; Mao et al. 2024) for rhythm and semantic synchronization, and BenchUFO (Wang and Yang

Table 3 Comparison of category-level datasets for diffusion-based video generation

Dataset	Year	Categories	Clips	Resolution	Domain
UCF-101 (Soomro et al. 2012)	2012	101	13K	320×240	Human Actions
Cityscapes (Cordts and et al. 2016)	2015	30	3K	256×256	Urban Driving
Moving MNIST (Srivastava and et al. 2015)	2016	10	10K	64×64	Digits (Synthetic)
BAIR (Ebert and et al. 2017)	2017	2	45K	64×64	Robot Arm
DAVIS (Perazzi and et al. 2016)	2017	–	90	1280×720	Object Segmentation
Kinetics-400 (Kay et al. 2017)	2017	400	260K	256×256	Human Actions
Sky Time-Lapse (Xiong et al. 2018)	2018	1	38K	256×256	Nature / Sky
Something-Something V2 (Goyal et al. 2017)	2018	174	220K	256×256	Human Actions
Epic-Kitchen (Damen and et al. 2018)	2018	149	90K	1920×1080	Egocentric Cooking
TaiChi-HD (Siarohin and et al. 2019)	2019	1	3K	256×256	Human Motion
BridgeData (Walke et al. 2024)	2021	10	7K	256×256	Robot Tasks
RDS (VideoLDM) (Liang and et al. 2022)	2023	2	683K	512×1024	Autonomous Driving

Table 4 Summary of evaluation metrics for diffusion-based video generation

Metric Type	Metric	Key Property
Image-level	FID (Heusel et al. 2017)	Visual realism (distribution distance between real and generated frames)
	PSNR (Hore and Ziou 2010)	Pixel-level reconstruction accuracy
	SSIM (Wang et al. 2004)	Structural similarity and visual coherence
	CLIPSim (Radford et al. 2021)	Frame-text semantic alignment using CLIP
Video-level	FVD (Unterthiner et al. 2019)	Temporal consistency and perceptual realism
	KVD (Ryan et al. 2023)	KL divergence-based distribution similarity
	Video IS (Wu et al. 2016)	Frame-level realism and intra-video diversity
	Frame Consistency (Wu et al. 2022)	Smoothness and coherence across adjacent frames
	VideoCLIPSim (Fang et al. 2023)	Video-caption alignment using VideoCLIP
Specialized	Motion Consistency (Liu et al. 2023)	Smooth and plausible motion transition
	AV-Align (Haji-Ali et al. 2024)	Audio-video semantic or rhythmic synchronization
	BenchUFO (Wang and Yang 2025)	LLM-based consistency and controllability evaluation

2025) for LLM-based assessment. These complement traditional metrics by incorporating human perception and multi-condition controllability.

Overall, a combination of frame-level accuracy, temporal smoothness, multimodal alignment, and diversity is essential for robust evaluation. Developing unified, scalable, and task-specific benchmarks remains a key direction for future research.

5 Performance comparison

This section presents a comprehensive comparison of recent text-to-video generation models evaluated on two widely used benchmarks: UCF-101 (Soomro et al. 2012) and MSR-VTT (Xu et al. 2016). The comparison is based on key quantitative metrics, including Fréchet Video Distance (FVD) (Unterthiner et al. 2019), Inception Score (IS) (Wu et al. 2016), Fréchet Inception Distance (FID) (Heusel et al. 2017), and CLIP Similarity (CLIPSim) (Radford et al. 2021), which jointly reflect the visual quality, motion coherence, and semantic consistency of generated videos.

The results in Table 5 are based on the pre-trained weights provided by the official repositories of each method or on the models trained using their released code. The best-performing results for each indicator are highlighted in bold within the table. In the method column of Table 5, the method with $*$ is the zero-shot method. At the same time, we also presented some visualizations from the original paper, as shown in Fig. 6, Fig. 7 and Fig. 8. We hope to use these visualizations to give readers a more intuitive understanding of this field. Figure 7 and 8 present visualizations from three leading open-source SOTA methods: SVD (Blattmann et al. 2023), CogVideoX (Yang et al. 2024), and Huanyuanvideo (Kong et al. 2024), these results illustrate the strengths of current SOTA models in semantic fidelity and temporal continuity, each exhibiting distinctive visual characteristics. The following presents our experimental analysis.

Table 5 Performance comparison of text-to-video generation methods on UCF-101 and MSR-VTT datasets

Method	Year	Data	Res.	UCF-101			MSR-VTT		
				FID(↓)	FVD(↓)	IS(↑)	FID(↓)	FVD(↓)	CLIPSim(↑)
TATS (Ge et al. 2022)	2022	–	256×256	–	332	**79.28**	–	–	–
MMVG∗ (Fu et al. 2023)	2022	–	128×128	–	395	58.3	60.6	–	0.2385
NUWA (Wu et al. 2022)	2022	–	256×256	–	–	–	–	–	0.2402
LVDM∗ (He et al. 2022)	2022	2 M	256×256	–	641.8	–	–	742	0.2381
Magic-Video∗ (Zhou et al. 2022)	2022	10 M	256×256	145.00	699.00	–	–	998	–
Make-A-Video∗ (Singer et al. 2022)	2022	273	256×256	–	367.23	33.00	13.17	–	0.3049
ED-T2V∗ (Liu et al. 2023)	2023	10 M	256×256	–	–	–	–	–	0.2763
InternVid∗ (Wang et al. 2024)	2023	28 M	256×256	60.25	616.51	21.04	–	–	0.2951
Video-LDM∗ (Blattmann et al. 2023)	2023	10 M	256×256	–	550.61	33.45	–	–	0.2929
Video-Composer∗ (Wang et al. 2023)	2023	10 M	256×256	–	–	–	–	580	0.2932
Latent-shift∗ (An et al. 2023)	2023	10 M	256×256	–	–	–	15.23	–	0.2773
Video-Fusion∗ (Luo et al. 2023)	2023	10 M	256×256	75.77	639.90	17.49	–	581	0.2795
Make-Your-Video∗ (Xing et al. 2024)	2023	10 M	256×256	–	330.49	–	–	–	–
PYoCo∗ (Ge et al. 2023)	2023	22.5M	256×256	–	355.19	47.76	**9.73**	–	–
CoDi∗ (Tang et al. 2023)	2023	-	512×512	–	–	–	–	–	0.2890
NExT-GPT∗ (Wu et al. 2024)	2023	-	320×576	–	–	–	13.04	–	0.3085
Dysen-VDM∗ (Fei et al. 2024)	2023	10 M	256×256	–	325.42	35.57	12.64	–	0.3204
VideoFactory∗ (Wang et al. 2025)	2023	-	256×256	–	410.00	–	–	–	0.3005
ModelScope∗ (Wang et al. 2023)	2023	10 M	256×256	–	410.00	–	11.09	550	0.2930
VideoGen∗ (Li et al. 2023)	2023	10 M	256×256	–	554.00	71.61	–	–	0.3127
Animate-A-Story∗ (He et al. 2023)	2023	10 M	256×256	–	515.15	–	–	–	–
VidRD∗ (Gu et al. 2023)	2023	5.3M	256×256	–	363.19	39.37	–	–	–
CogVideo (Chinese)∗ (Hong et al. 2022)	2023	–	480×480	–	751.34	23.55	–	–	0.2614
CogVideo (English)∗ (Hong et al. 2022)	2023	–	480×480	–	701.59	25.27	–	–	0.2631
LAVIE∗ (Wang et al. 2024)	2023	10 M+25 M	320×512	–	526.30	–	–	–	0.2949
VideoDirGPT∗ (Lin et al. 2023)	2023	10 M	256×256	–	–	–	12.22	550	0.2860
Show-1∗ (Zhang et al. 2024)	2023	10 M	320×576	–	394.46	35.42	13.08	538	0.3072

Table 5 (continued)

Method	Year	Data	Res.	UCF-101			MSR-VTT		
				FID(↓)	FVD(↓)	IS(↑)	FID(↓)	FVD(↓)	CLIPSim(↑)
Dynamicrafter∗ (Xing et al. 2024)	2023	10 M	256×256	–	429.23	–	–	234	–
EMU-Video∗ (Girdhar et al. 2024)	2023	10 M	256×256	–	606.20	42.70	–	–	–
SVD∗ (Blattmann et al. 2023)	2023	577 M	256×256	–	**242.02**	–	–	–	–
W.A.L.T∗ (Gupta et al. 2024)	2023	89 M	128×128	–	258.1	35.1	–	244.7	–
CMD∗ (Yu et al. 2024)	2023	10 M	512×1024	–	504.00	–	–	–	0.2894
ModelScopeT2∗(Wang et al. 2023)	2023	-	32×32	–	–	–	11.9	550	0.2930
Lumiere∗ (Bar-Tal et al. 2024)	2024	10 M	320×576	–	332.49	37.54	–	–	–
RF (Long et al. 2024)	2024	-	320×512	–	–	–	10.8	153	0.2859
ART●V∗ (Weng et al. 2024)	2024	10 M	320×320	–	315.69	50.34	–	291	0.2859
AnimateDiff (Liu et al. 2023)	2024	10 M	254×254	–	499.3	–	-	–	0.2880
MicroCinema∗ (Wang et al. 2024)	2024	10 M	256×256	–	342.86	37.46	–	377	0.2967
Vlogger∗ (Zhuang et al. 2024)	2024	10K	320×512	–	292.43	–	–	–	0.2908
Make Pixels Dance∗ (Zeng et al. 2024)	2024	10 M	256×256	49.36	242.82	42.10	–	381	0.3125
SimDA∗ (Xing et al. 2024)	2024	10 M	256×256	–	–	–	–	456	0.2945
Snap Video∗ (Menapace et al. 2024)	2024	10 M	288×288	**39.0**	260.1	38.89	–	**110.4**	0.2793
MoVideo∗ (Liang et al. 2024)	2024	–	256×256	–	313.41	34.13	12.71	–	**0.3213**
MagDiff∗ (Zhao et al. 2024b)	2024	5.3M+76K	256×256	–	339.62	48.57	–	245	–
HARIVO (Kwon et al. 2024)	2024	–	256×256	–	–	–	–	787.87	0.2948
VersVideo-L∗ (Xiang et al. 2023)	2024	–	256×256	–	–	72.9	–	207	–

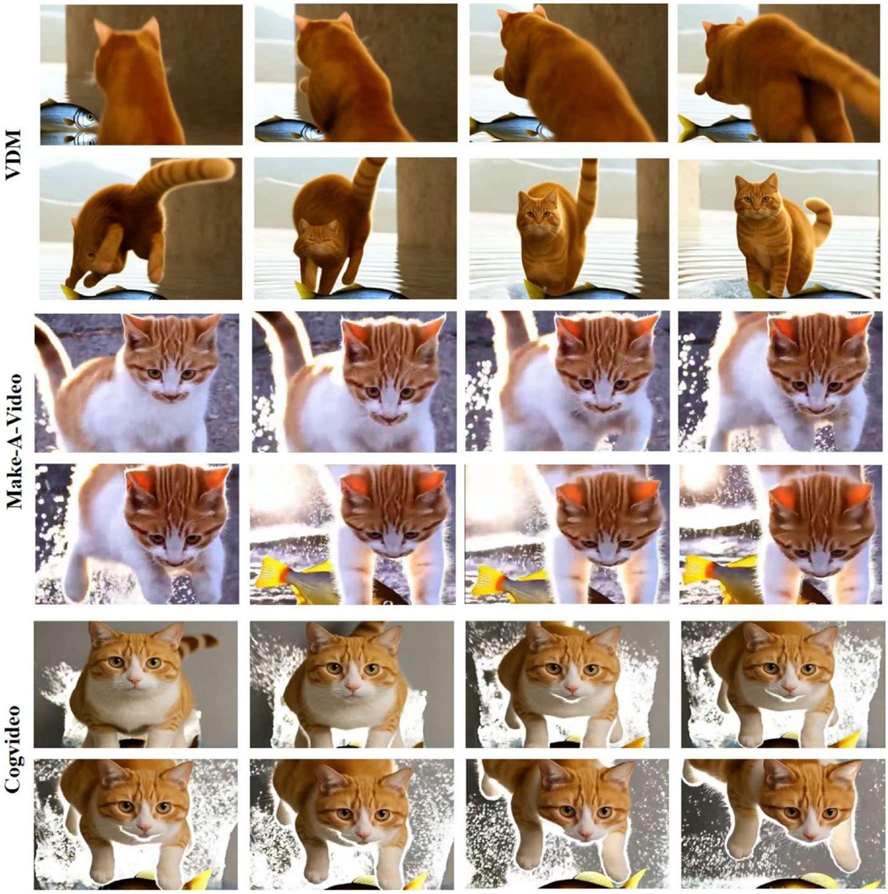

Fig. 6 Comparison between VDM (Ho et al. 2022) (top), Make-A-Video (Singer et al. 2022) (middle), and CogVideo (Hong et al. 2022) (bottom) for the prompt "An orange cat jumped up and pounced on the fish in the water". Representative frames are shown from our implementation of the models, using official codebases and pre-trained weights

5.1 Comparative analysis on UCF-101 and MSR-VIT

UCF-101 is widely adopted to evaluate motion realism and dynamic consistency in generated videos. As observed in Table 5, recent methods trained on large-scale datasets tend to perform better in terms of Fréchet Video Distance (FVD) and Inception Score (IS). Notably, Snap Video (Menapace et al. 2024) achieves the best FID of 39.00 and a low FVD of 260.1, along with an IS of 38.89, showing its strong capability in generating temporally coherent and visually rich videos.

PixelDance (Zeng et al. 2024) also demonstrates excellent motion quality with an FVD of 242.82 and IS of 42.10. MoVideo (Liang et al. 2024) balances both metrics well, reaching 313.41 (FVD) and 34.13 (IS). SVD (Blattmann et al. 2023) reports the best FVD (242.02) among all, reflecting the benefit of training on large-scale datasets (577 M samples).

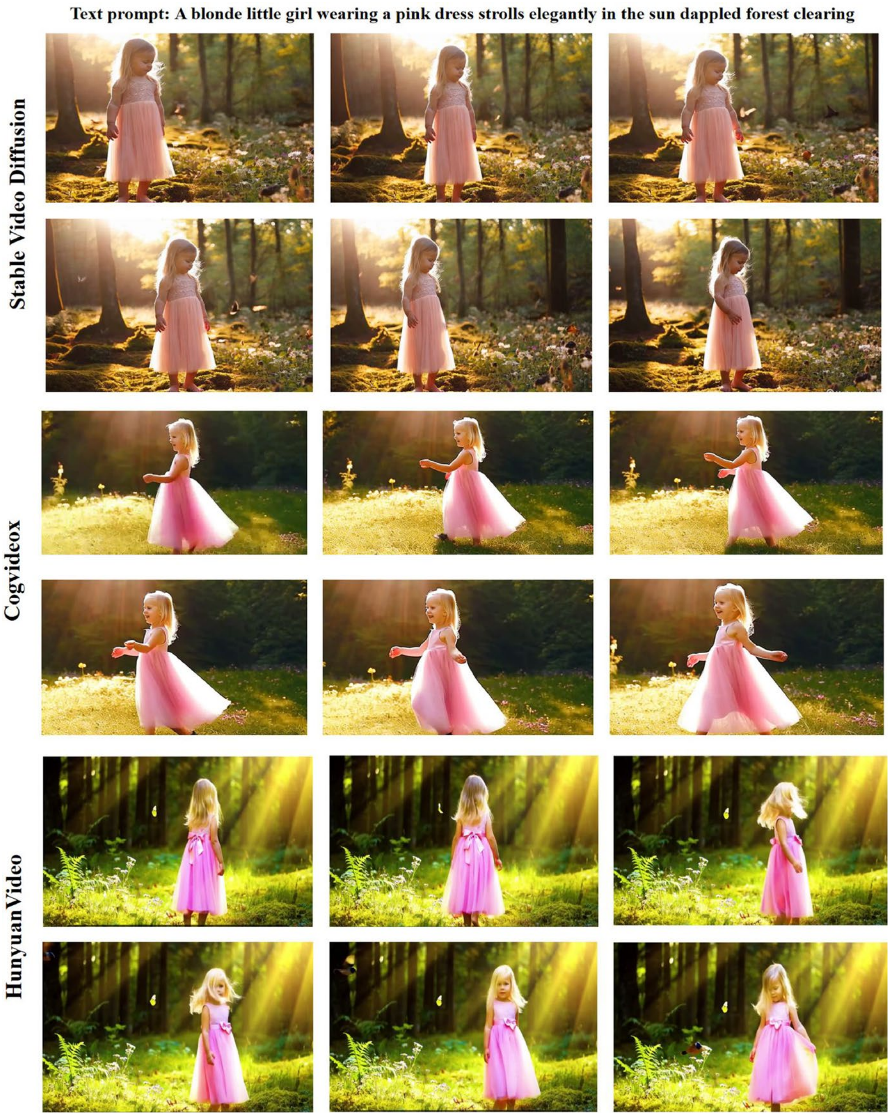

Fig. 7 A comparison of different SOTA diffusion video generation models using pre-trained weights for multi-frame visualization. From top to bottom: SVD (Blattmann et al. 2023), CogVideoX (Yang et al. 2024), and Huanyuanvideo (Kong et al. 2024)

Older methods such as LVDM (He et al. 2022) and Magic-Video (Zhou et al. 2022) exhibit relatively higher FVDs (641.8 and 699.0, respectively), indicating the significant improvement over time brought by architectural advances and training scale.

MSR-VTT is the primary benchmark for evaluating text-video semantic alignment, with CLIPSim as the core metric. As seen in Table 5, MoVideo (Liang et al. 2024) leads with a CLIPSim of 0.3213, followed closely by Dysen-VDM (Fei et al. 2024) (0.3204), Pixel-

Fig. 8 Same to Fig. 7

Dance (Zeng et al. 2024) (0.3125), and VideoGen (Li et al. 2023) (0.3127), all demonstrating robust multimodal alignment.

NExT-GPT (Wu et al. 2024) and Show-1 (Zhang et al. 2024) achieve CLIPSim scores over 0.30 while also maintaining low FVDs. These models leverage large-scale video-language pretraining and diffusion-transformer architectures to effectively align textual prompts with visual content.

In contrast, models like CogVideo (Hong et al. 2022) (English version: 0.2631) and NUWA (Wu et al. 2022) (0.2402) show lower CLIPSim scores, reflecting earlier-stage methods that lacked refined cross-modal attention mechanisms.

5.2 Analysis of SOTA model performance

Recent SOTA models in diffusion-based video generation demonstrate substantial improvements in both visual fidelity and semantic alignment. In terms of model architecture, Snap Video adopts a multi-stage diffusion process that decouples motion modeling from frame synthesis, effectively mitigating temporal error accumulation. In its global local feature extraction mechanism, global features maintain scene consistency, while local features focus on optimizing dynamic details. This explicitly modular architecture contributes significantly to the model's superior motion smoothness.

PixelDance achieves its performance gains through an innovative local attention mechanism that explicitly constrains fine-grained motion correlations between adjacent frames. This allows the model to better capture subtle motion dynamics. Additionally, its multi-scale context fusion facilitates temporal dependency modeling across different time scales, supporting strong temporal consistency in long video sequences.

The impact of training data size on model performance is particularly significant. Taking SVD as an example, its 577 million sample training set provides the model with rich spatio-temporal distribution priors, which enables the model to infer more reasonable time transitions when generating long videos. In contrast, models trained on smaller scale data, such as VideoGen, often require a compromise between text alignment and perceptual quality.

In terms of cross modal understanding, MoVideo achieves strong semantic alignment through a large-scale text-video pretraining strategy using a cross-modal Transformer architecture. This design enables accurate interpretation of spatiotemporal concepts described in text, which are then transformed into coherent and semantically rich visual content.

The current evaluation system has several important limitations: firstly, different indicators focus on different aspects, with FVD placing more emphasis on motion quality and IS leaning towards static image quality, which may lead to conflicts in model optimization objectives. Secondly, the difference in output resolution (ranging from 128×128 to 512×1024) directly affects the evaluation of perceptual quality, but this factor is often overlooked in existing benchmark tests. Finally, some methods (such as Animate-A-Story) lack complete evaluation results, further complicating direct comparisons across models.

These performance variations reflect differing design priorities among research teams: some (e.g., SVD) emphasize scalability, while others (e.g., PixelDance) prioritize precise modeling of motion dynamics. This diversity highlights the complexity of video generation and underscores the need for unified and standardized evaluation criteria.

5.3 Summary

In summary, Table 5 reflects the rapid advancement of diffusion-based text-to-video models, with clear improvements in motion smoothness, image fidelity, and semantic alignment. The top-performing models such as Snap Video, MoVideo, and PixelDance showcase the potential of large-scale diffusion learning combined with advanced architectural innovations. Future benchmarking should focus on unified metric reporting, resolution normalization, and task-specific evaluation to ensure fairer comparisons across the diverse landscape of video generation models.

6 Solved and unsolved challenges

6.1 Overview of achievements

In recent years, diffusion-based video generation has made remarkable progress, driving both theoretical innovation and practical application. Key achievements include:

- **Enhanced temporal coherence:** Multi-stage refinement networks and dynamic temporal alignment modules have substantially improved frame-to-frame continuity, reducing flickering and inconsistency in generated videos.
- **Improved semantic fidelity:** Integration of multimodal conditions such as text, pose, audio, and depth maps has enabled models to generate videos that align closely with user-specified prompts. Approaches leveraging cross-attention and latent conditioning, exemplified by Snap Video(Menapace et al. 2024), PixelDance(Zeng et al. 2024), and MoVideo(Liang et al. 2024), have demonstrated fine-grained semantic control.
- **Efficiency gains:** Latent diffusion models (e.g., LDM(Rombach et al. 2022)) and adaptive sampling schedules have reduced inference time and computational load, making high-quality generation more practical.
- **Advanced architectures:** The adoption of diffusion-transformer hybrids and temporal attention mechanisms has improved both global scene understanding and local motion consistency, supporting complex tasks such as story-driven generation and dynamic scene synthesis.
- **New evaluation metrics:** While FVD(Unterthiner et al. 2019) and IS(Wu et al. 2016) remain foundational, the field now incorporates metrics like CLIPSim(Radford et al. 2021) to address previously overlooked dimensions, particularly the semantic fidelity of generated content to input prompts.

6.2 Key unsolved challenges

Despite the progress, the field faces several unsolved challenges:

- **Temporal consistency in long videos:** Existing methods often exhibit drift or incoherence in long sequences, requiring explicit memory or hierarchical modeling.
- **Cross-modal integration limitations:** Ensuring consistent alignment across diverse modalities (text, audio, pose) is challenging, especially for complex prompts.
- **Graph Diffusion Models:** Current diffusion models underutilize graph structures, which could improve scene coherence and structural reasoning.
- **Inference efficiency:** High computational costs remain a bottleneck for practical deployment, particularly for high-resolution or real-time applications.
- **Evaluation consistency:** Inconsistent metrics and varying output resolutions hinder fair benchmarking, calling for standardized protocols.

These unresolved issues highlight the need for further research into temporal modeling, efficient architectures, cross-modal fusion, and comprehensive evaluation strategies. A more detailed discussion of these challenges, along with corresponding future research directions, is provided in SectionÂ 7.

7 Future research directions and open problems

This section synthesizes key trends and emerging opportunities in diffusion-based video generation. We summarize unresolved issues, promising future directions, and emerging technologies that may drive the next stage of research.

7.1 Enhancing temporal coherence and efficiency

Ensuring temporal stability remains one of the most persistent challenges in video generation. While short videos can achieve frame-level coherence through simple spatial constraints, long-form synthesis often suffers from temporal drift and motion inconsistency. Future research should focus on memory-augmented architectures, recurrent latent dynamics, and hierarchical generation strategies that explicitly capture long-range dependencies. Techniques such as temporal recycling, latent interpolation, and causal conditioning can be employed to maintain context continuity in multi-second or minute-long videos. In particular, incorporating memory mechanisms (e.g., recurrent latent states) and structured priors (e.g., scene graphs) can significantly improve long-range temporal reasoning and visual consistency.

The growing complexity of diffusion models also presents challenges for inference speed, scalability, and resource efficiency. While multi-stage and cascaded frameworks (e.g., Show-1 (Zhang et al. 2024), FlexiFilm (Ouyang et al. 2024)) offer enhanced quality, they come at the cost of computationally intensive sampling. Future directions include designing lightweight backbones (e.g., sparse transformers, hybrid UNet, ViT), adaptive sampling schedules, and plug-in modules (e.g., adapters, dynamic denoising paths) to balance quality with speed. Furthermore, model distillation techniques offer a promising way to reduce the inference burden by transferring knowledge from large diffusion models into compact, efficient variants suitable for real-time or edge deployment.

7.2 Advancing cross-modal integration and graph-based reasoning

Recent methods have demonstrated the effectiveness of multimodal guidance (e.g., text, audio, pose, depth, brain activity), but maintaining semantic consistency across these diverse inputs remains challenging. Future research should explore unified embedding spaces, hierarchical conditioning, and token-level cross-modal fusion for better integration. Applications such as sound-guided motion (e.g., lip-sync or dance) and layout-driven synthesis can benefit from stronger cross-modal alignment.

Graph-based diffusion models present a promising yet underexplored direction. By modeling scene structures and temporal relationships as graphs, these approaches can enhance scene coherence, multi-object interactions, and complex event synthesis. Integrating graph structures into diffusion frameworks could significantly improve structural reasoning and control in long, dynamic sequences.

Specifically, graph diffusion models can represent relationships between entities such as objects, motions, or semantic regions, enabling more fine-grained control and interpretability. Unlike traditional token-based methods, they allow explicit modeling of spatiotemporal dependencies and interactions, thus improving temporal consistency in long videos. Leveraging graph neural networks (GNNs) within diffusion frameworks can also facilitate structured learning of complex scenes, supporting scenarios with intricate motions, multi-object dynamics, and enhanced scene coherence.

7.3 Improving generalization and control flexibility

Diffusion models trained on fixed domains often struggle with generalization to open-domain scenarios or unseen combinations of prompts. Future work should focus on data-efficient pretraining, unsupervised learning, and multi-domain adaptation strategies. Few-shot or zero-shot models leveraging vision-language foundation models (VLFMs) offer potential for improving generalization while minimizing retraining requirements.

Strong conditioning signals, such as fixed images or depth maps, can lead to overfitting or suppression of motion diversity. Adaptive conditioning strength modulation, semantic filtering of control inputs, and multi-scale conditioning refinement are essential for maintaining control precision while preserving diversity, particularly for tasks like image-to-video, depth-driven editing, and audio-visual synthesis.

7.4 Towards unified frameworks and human-centered interaction

Currently, video generation tasks are often handled in isolation, with separate models optimized for specific tasks. A promising direction is to develop unified frameworks that support flexible task switching and modular component reuse, such as multi-head transformers, hierarchical decoders, and routing networks. These can enable comprehensive applications like instruction-based video editing or scene-aware generation with multi-resolution outputs.

Human-interactive, real-time generation systems are increasingly needed for practical deployment. Future research should focus on intuitive interfaces, such as sketching, drag-based motion cues, and mixed-modality prompts, alongside progressive refinement and conditional feedback mechanisms. Low-latency pipelines with explainable intermediate states will be crucial for creative and educational applications.

7.5 Ethical considerations and broad applications

The rise of diffusion-based video generation brings ethical concerns about misinformation, privacy, and bias. Future research must focus on developing safety filters, bias mitigation techniques, and transparent generation pipelines. Standardized evaluation protocols assessing visual quality, temporal coherence, semantic alignment, and realism are needed for fair comparison and trustworthy deployment.

Applications of diffusion-based video generation are expanding across industries, including lifelike character animation, virtual avatars, film effects, immersive VR/AR simulations, dynamic medical video synthesis, and personalized media creation. Further potential lies in educational technology, environmental monitoring, and creative content generation, demonstrating the broad societal impact of this technology.

Overall, future research on diffusion-based video generation must continue to address several key challenges, including improving generation efficiency, reducing computational costs, enhancing long-term temporal stability, and developing unified evaluation metrics. Successfully overcoming these barriers will further boost the applicability of diffusion models in a wide range of real-world scenarios, such as film production, autonomous driving, innovative healthcare, digital humans, and immersive media.

8 Conclusions

In summary, diffusion-based video generation has made remarkable strides in both academic research and practical applications. These models have demonstrated superior capabilities in generating high-fidelity, temporally coherent, and semantically controllable videos, often outperforming traditional generative approaches such as GANs(Dhariwal and Nichol 2021) and autoregressive models. Key advances include the incorporation of sophisticated architectures (e.g., UNet(Çiçek et al. 2016), Transformers(Yan et al. 2021)), adaptive sampling strategies, and effective multimodal conditioning (e.g., text, pose, audio), which have enabled the generation of videos with enhanced realism and control.

This review has highlighted the strengths and limitations of state-of-the-art diffusion-based video generation methods, offering a systematic analysis of their design principles, evaluation metrics, and benchmark performance. Our analysis shows that while recent state-of-the-art models achieve impressive results in both perceptual quality and semantic alignment, challenges remain in terms of temporal consistency over long sequences, generalization to diverse domains, and computational efficiency.

Looking ahead, we observe several key trends and promising research directions. Future efforts should focus on improving long-term temporal modeling, enabling more efficient and scalable architectures, and advancing cross-modal integration for complex and controllable video synthesis. At the same time, architectures incorporating memory mechanisms (e.g., recurrent latent states) or structured priors (e.g., scene graphs) could unlock long-duration generation, while distillation techniques may enable real-time synthesis. Crucially, the field must adopt rigorous evaluation standards—separating resolution-dependent effects from true algorithmic progress through multi-scale benchmarks. Furthermore, standardized benchmarking protocols, perceptual metrics, and fair evaluation practices are essential for meaningful comparisons and trustworthy deployment of these models.

Overall, we believe that diffusion-based video generation is poised to revolutionize the field of generative visual media. This review provides a comprehensive foundation for future research, and we hope it inspires further exploration and innovation in this dynamic and impactful area.

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Data availability The experimental datasets supporting the results of this study are as follows, all of which are downloadable public datasets that can be used and downloaded from the corresponding website below: (1)UCF-101: https://www.crcv.ucf.edu/data/UCF101.php (2) MSR-VTT: https://paperswithcode.com/dataset/msr-vtt

Declarations

Conflict of interest The authors declare no Conflict of interest.

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